LEVERAGING CLOUDFRONT FOR HIGH-PERFORMANCE STATIC WEBSITE HOSTING WHETHER WITH AMAZON S3 OR INDEPENDENTLY

Overview:

To launch a static website without using Amazon S3 via CloudFront, EC2 and ELB services and to launch a website using Amazon S3 bucket via Cloudfront in the AWS console .

Description:

Case 1:

- Using **Amazon EC2 instances** as the hosting infrastructure, **Amazon CloudFront** for content distribution, and an **Elastic Load Balancer (ELB)** for dividing up incoming traffic across several EC2 instances comprise the stages involved in launching a website without utilizing Amazon S3.

Case 2:

- Using **Amazon S3** for storing the files which are used to host a static website(Amazon S3 is used to store the static files (HTML, CSS, JavaScript, images, etc.) and **Amazon CloudFront** for content distribution.

Prerequisites:

We must have some basic knowledge on:

1. Amazon EC2 (Elastic Compute Cloud) - Web service which provides scalable computing capacity in the cloud.

Instance - Virtual computing environments.

2. Amazon CloudFront - It is an **Content Delivery Network (CDN)** service, which is provided by Amazon Web Services, which helps to distribute content with low latency and high data transfer speeds, which improves application performance.

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3. Elastic Load Balancing – It is a service which is provided by Amazon in which the **incoming traffic is efficiently distributed across a group of servers** to increase the speed and performance of those servers.

- **Classic Load Balancers -** An older type of load balancer works on **Application Layer-7** and **Transport Layer-4** of Open System Interconnection (OSI) model.

- **Application Load Balancers -** An updated version of classic load balancer also called as version 2, which works on **Application Layer-7** of Open System Interconnection (OSI) model.

  Layer-7 accepts the protocols of **HTTP/HTTPS**.

- **Network Load Balancers –** A type of load balancer which works on **Transport Layer-4** of Open System Interconnection (OSI) model.

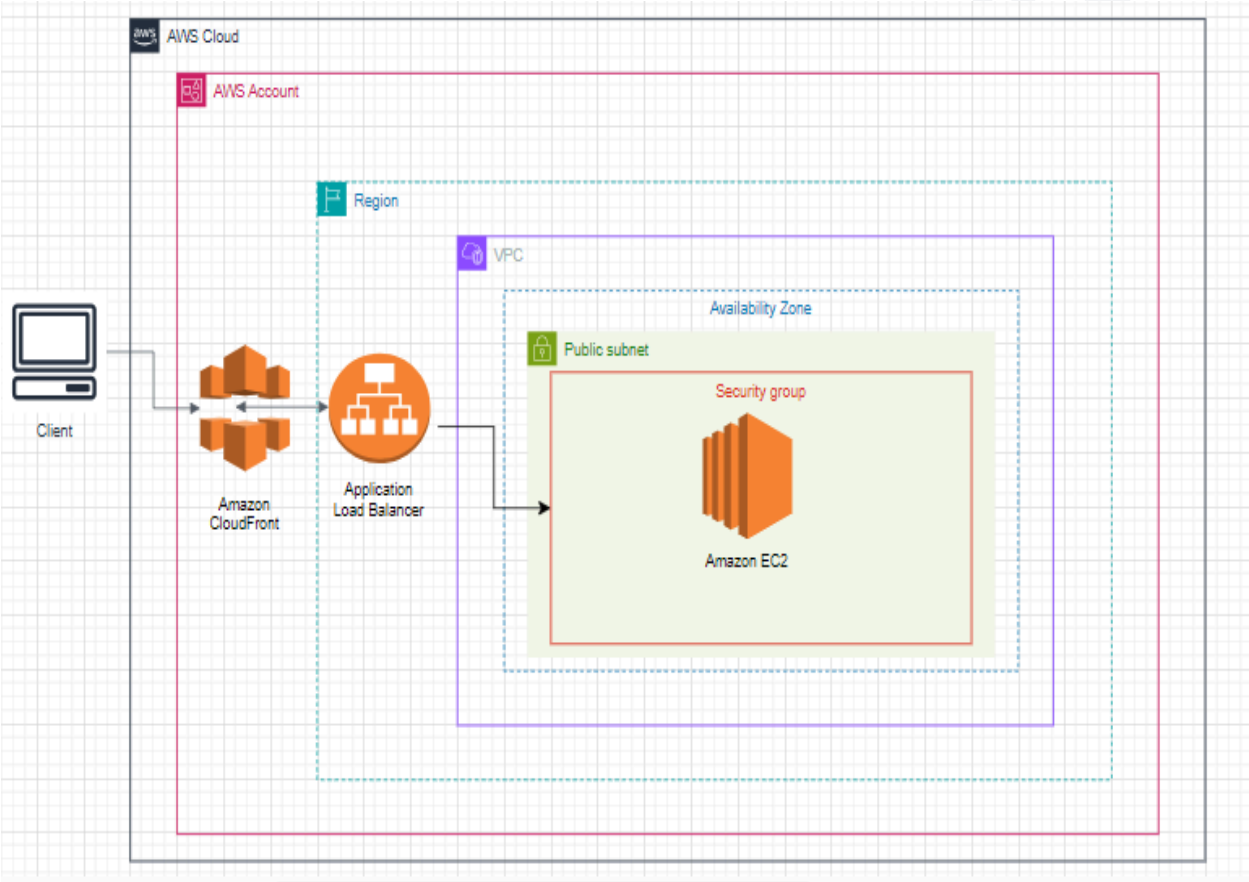
  Layer-4 accepts the protocols of **TCP/UDP**.
  When an application requires traffic with high performance and low latency then we use Application Load Balancing.

- **Gateway Load Balancers -** A type of load balancer enables you to deploy, scale, and manage virtual appliances in Amazon Web Services such as, Firewalls, Intrusion Detection and prevention systems, Deep packet inspection systems at your network layer.
  Gateway Load Balancer works on **Network Layer-3** of Open System Interconnection (OSI) model.

4. Amazon S3(Simple Storage Service) – Amazon S3 which is a simple storage service which is provided by Amazon Web Service which is a Cloud service provider, S3 is a Scalable storage service which is object level storage that offers scalability, data availability, security, and performance. We can use Amazon S3 to store and retrieve any amount of data at any time, from anywhere.

Case 1: Using **Amazon EC2 instances** as the hosting infrastructure, **Amazon CloudFront** for content distribution, and an **Elastic Load Balancer (ELB)** for dividing up incoming traffic across several EC2 instances comprise the stages involved in launching a website without utilizing Amazon S3.

Architecture:



Data Flow:

a. The client (such as a web browser) sends a request to access the website by entering the website's domain name into the browser's address bar (URL).

b. The request is first routed to the nearest CloudFront edge location. CloudFront acts as a Content Delivery Network (CDN), caching content and serving it from edge locations to reduce latency and improve performance.

c. If the requested content is not cached at the edge location or requires dynamic processing, CloudFront forwards the request to the designated origin, which in this case is the Application Load Balancer (ALB).

d. The ALB receives the forwarded request from CloudFront. It acts as the entry point for all incoming web traffic and distributes the request across multiple EC2 instances registered with the load balancer.

e. The ALB is associated with a security group that controls inbound and outbound traffic. The security group ensures that only authorized traffic is allowed to reach the EC2 instances behind the ALB.

f. The ALB forwards the request to one of the EC2 instances in the backend pool. These EC2 instances host the application logic and serve the requested content.

g. The EC2 instance processes the request and generates a response. If the request is for static content (e.g., HTML, CSS, JavaScript files), the EC2 instance directly serves the content. If the request requires dynamic content generation (e.g., querying a database), the EC2 instance executes the necessary application logic.

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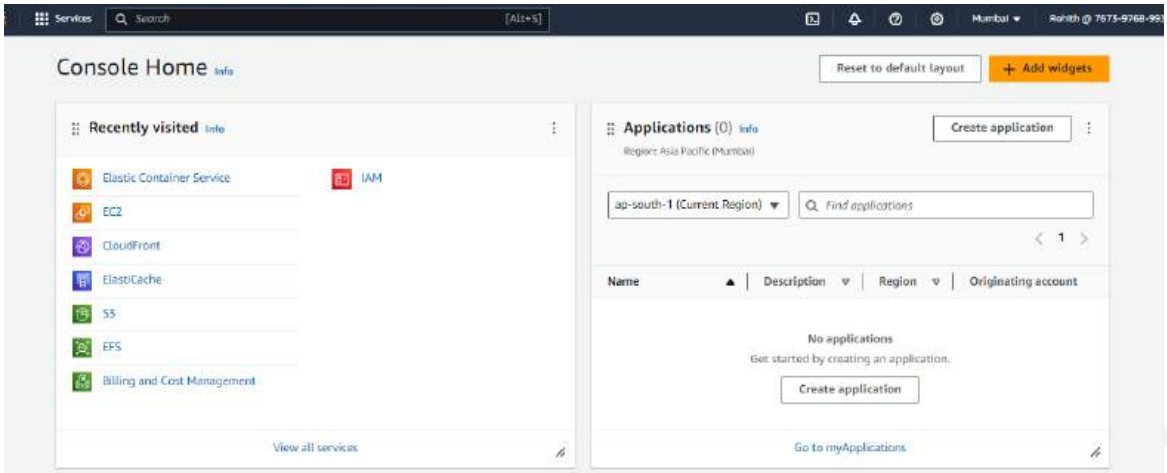
h. The response generated by the EC2 instance is sent back through the ALB to CloudFront. CloudFront caches the response (if applicable) and delivers it to the client.

i. Finally, the client receives the response from CloudFront and renders the website content in the web browser, completing the flow of the architecture.

Steps:

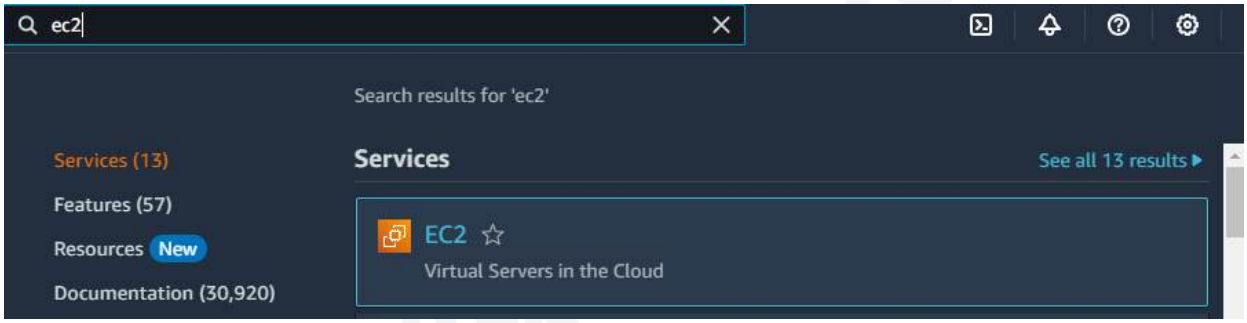
Step 1: Login into AWS Console

1.1 Create an account and login into it by providing the credentials.

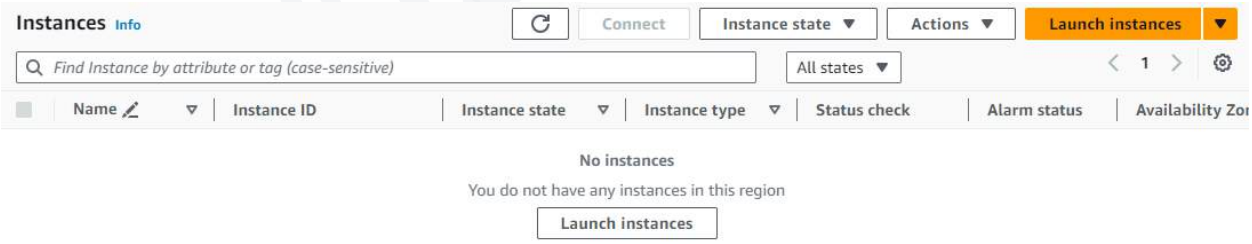


Step 2: Launch an EC2 Instance in Console

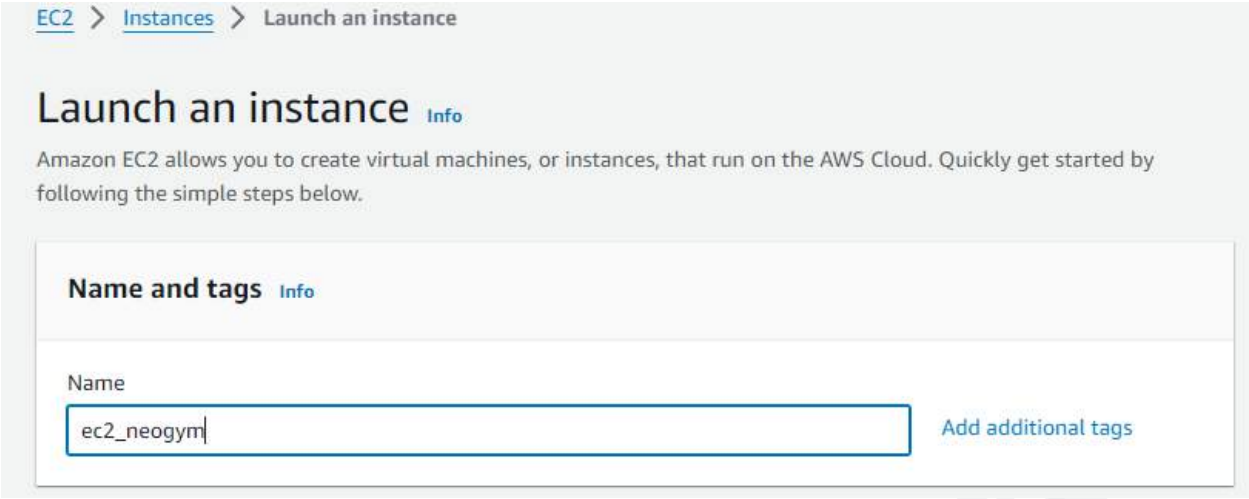
2.1 In Dashboard, search for **EC2 service,** click on **EC2** to open it.



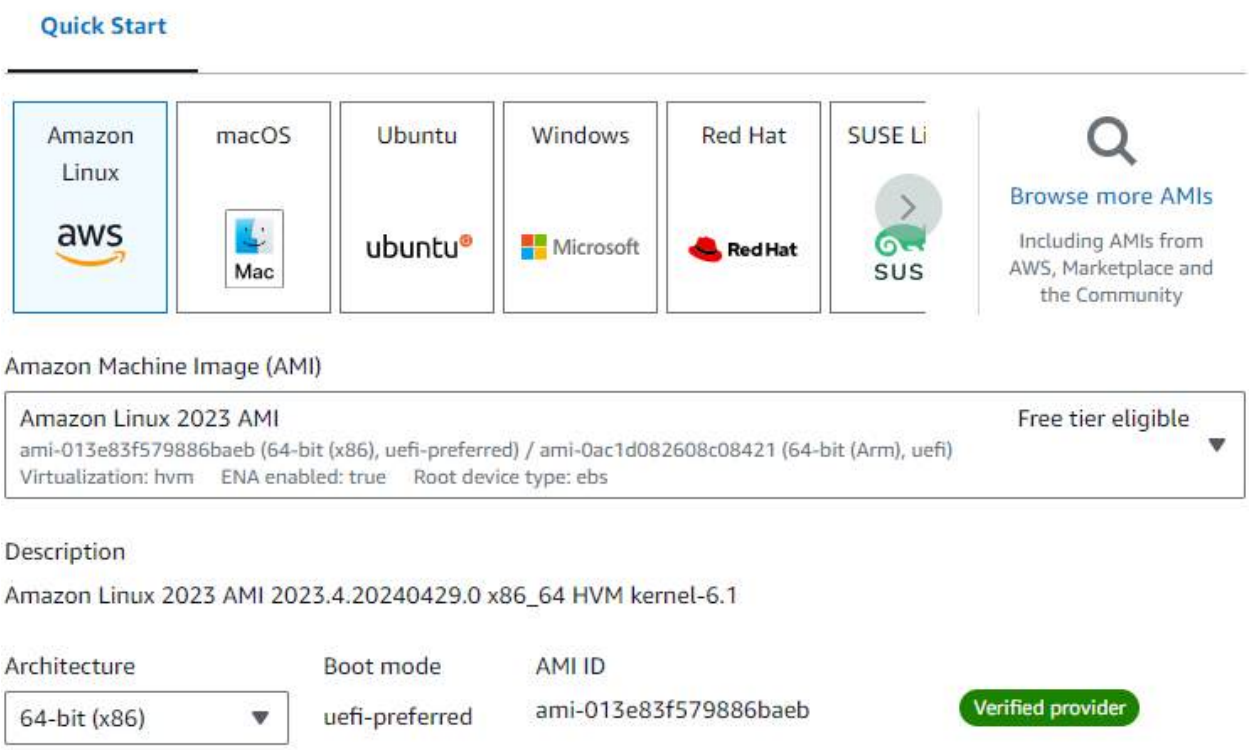
2.2 Click on **Launch instance** to configure.



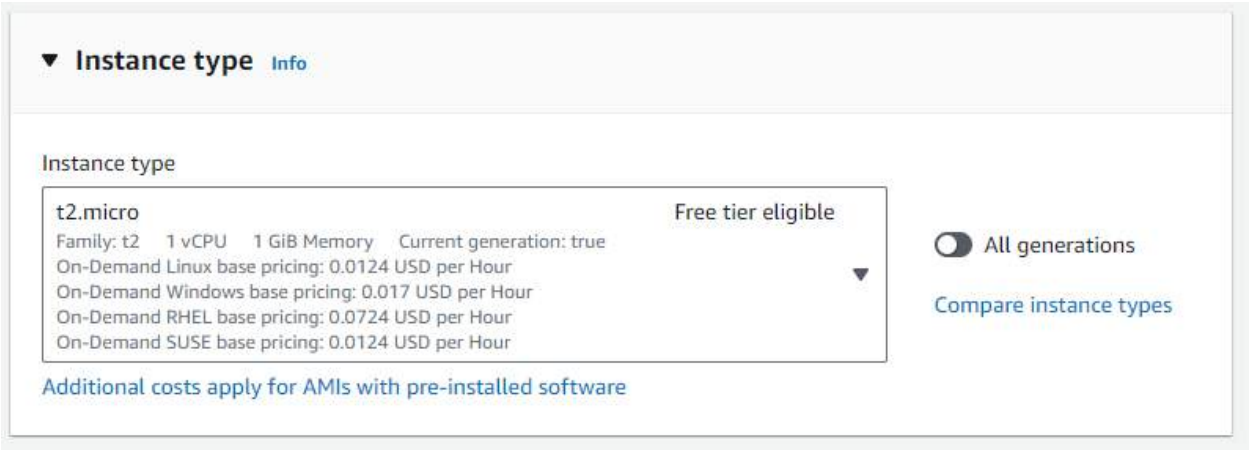
2.3 Type the **Name** which you prefer for naming the instance.



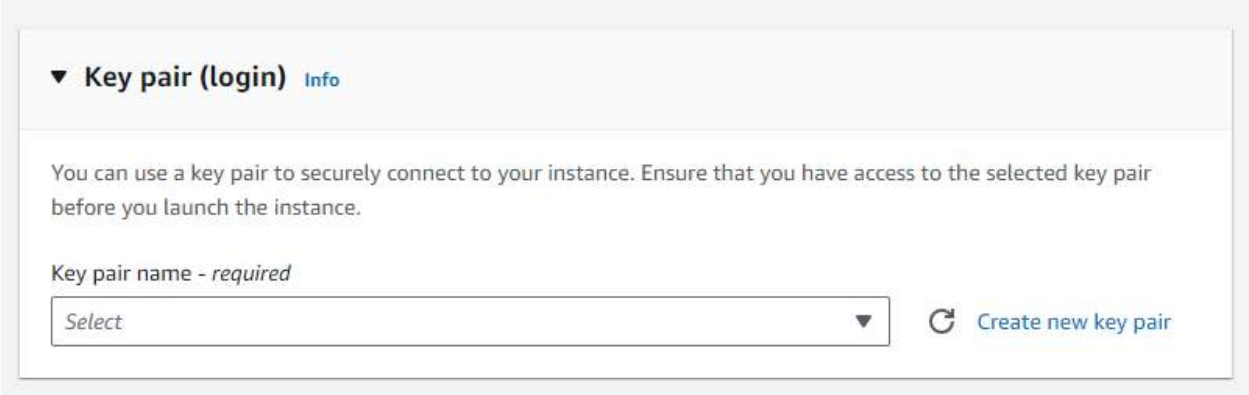
2.4 Choose the **AMI**, so here we are using **Linux**.



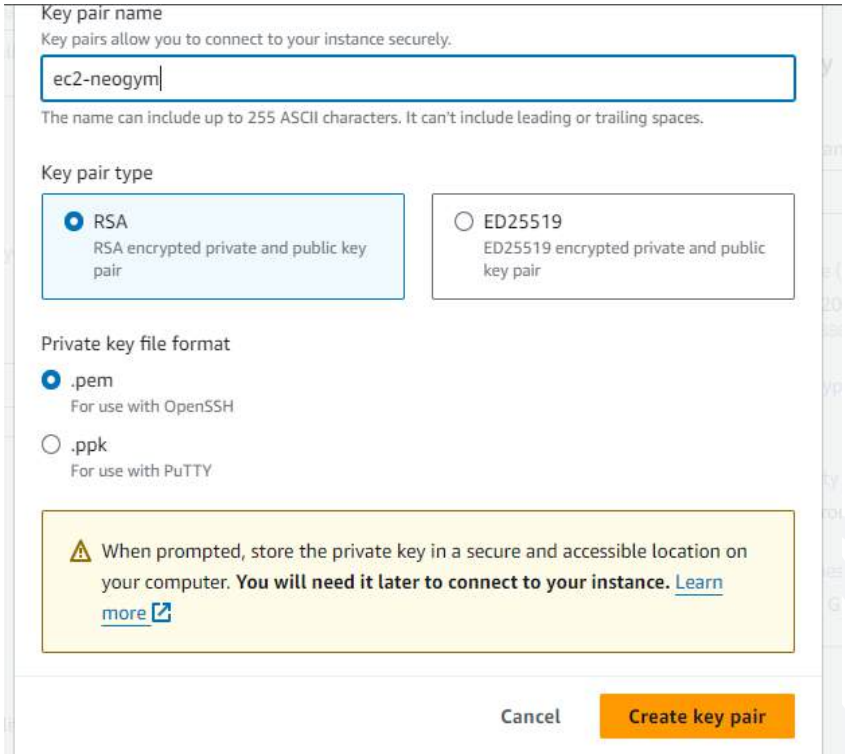
2.5 Select the instance type as **t2.micro**



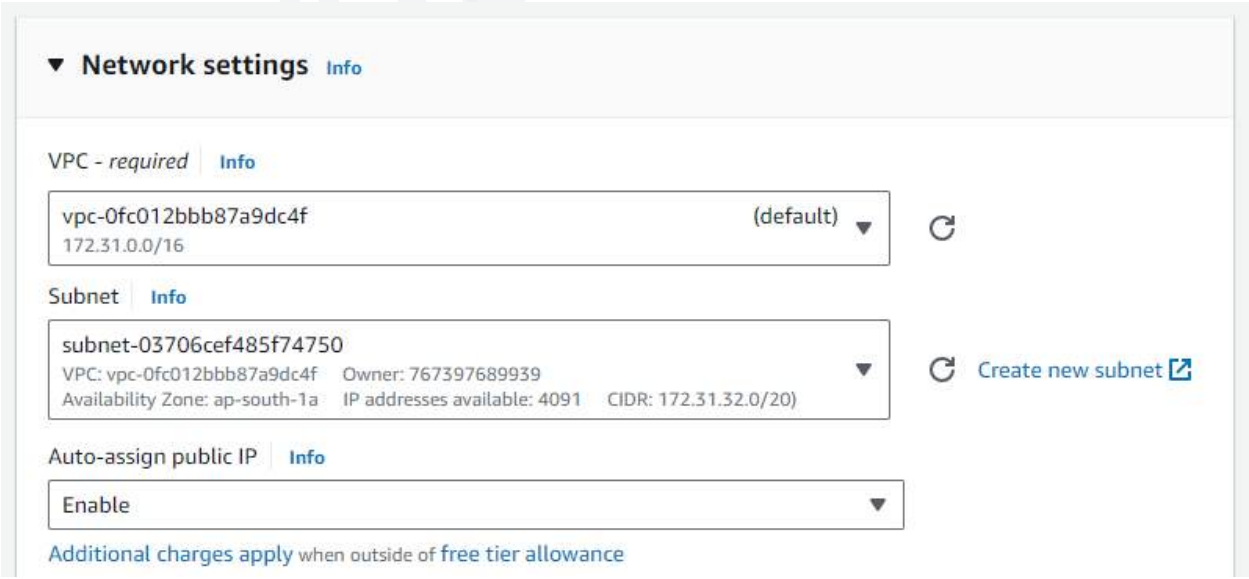
2.6 Create the **key Pair** to do that, Click on Create the **New pair key** for secure connection.



2.7 By clicking the **Create new pair key** a modal will pop up write the **key pair name** as you wish then under **key pair type** choose **RSA** radio button and after that **private key file format** should be considered. **pem** for OpenSSH. and then click on **Create key pair**.



2.8 Under Network Settings, click **Edit** and then assign a **Default VPC** or you can also assign a VPC which is created by you.



2.9 Create a **security group** option and then check inbound security group rules that the Type field is **HTTP** and source type field is **Anywhere**. If the above fields are not present then click **Add security group rule** and click those options.

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group | Select existing security group

Security group name - *required*

SG-NG

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and ._-:/()#,@[]+=&;{}!$*

Description - *required* Info

launch-wizard-1 created 2024-04-30T03:53:57.737Z

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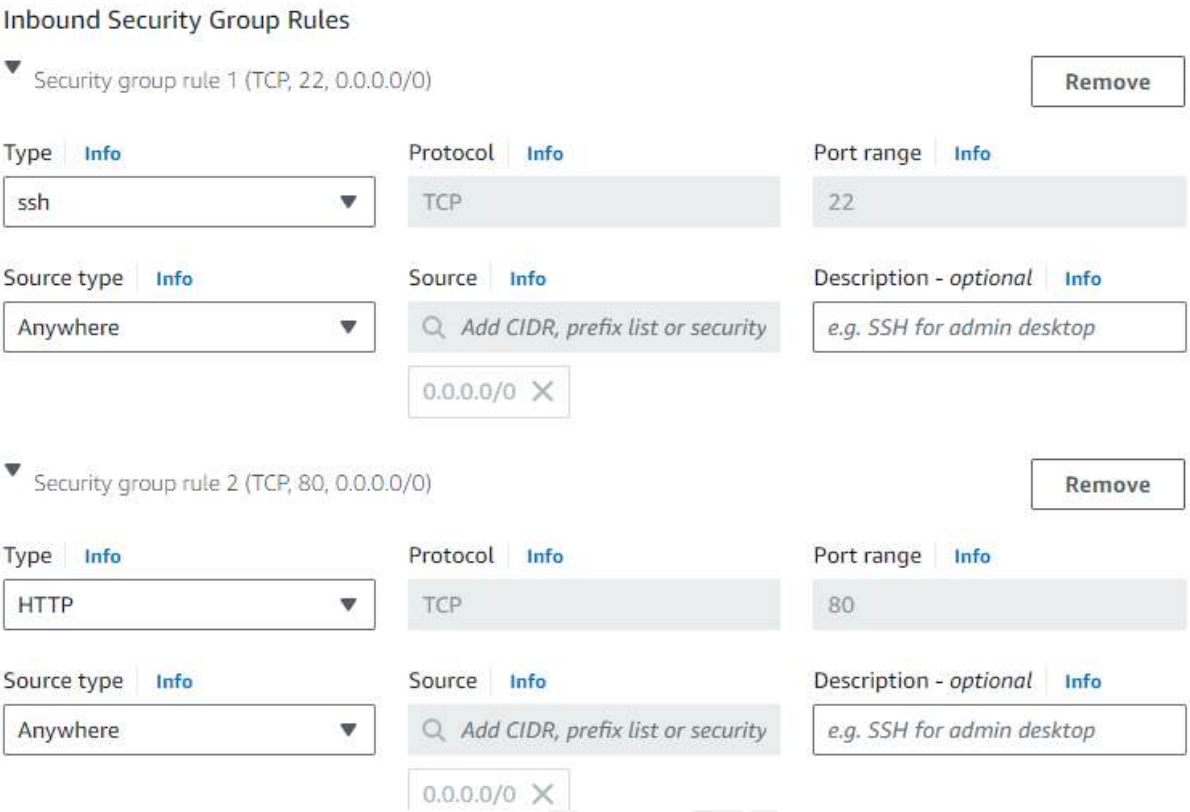
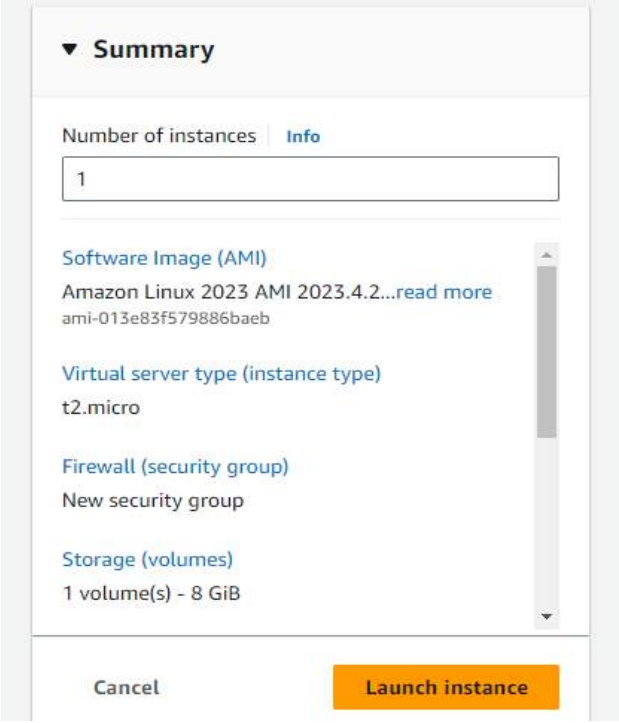
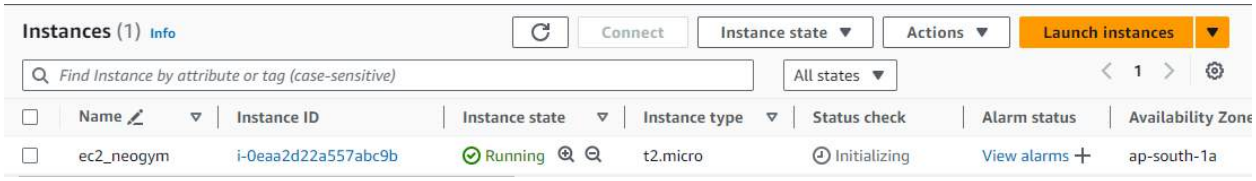


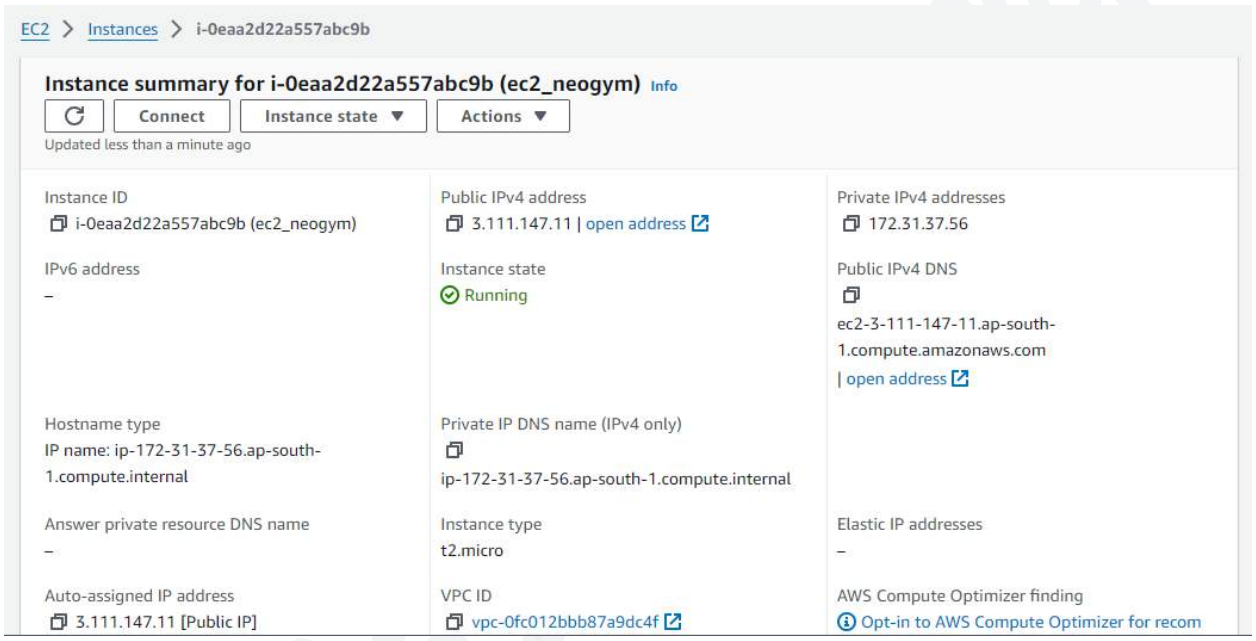
2.10 Leave the rest of the configuration as default and then click on **Launch Instance**.



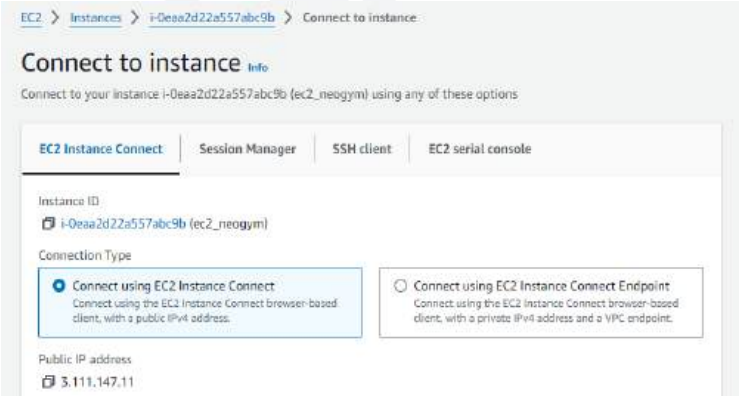
2.11 Instance is successfully created.

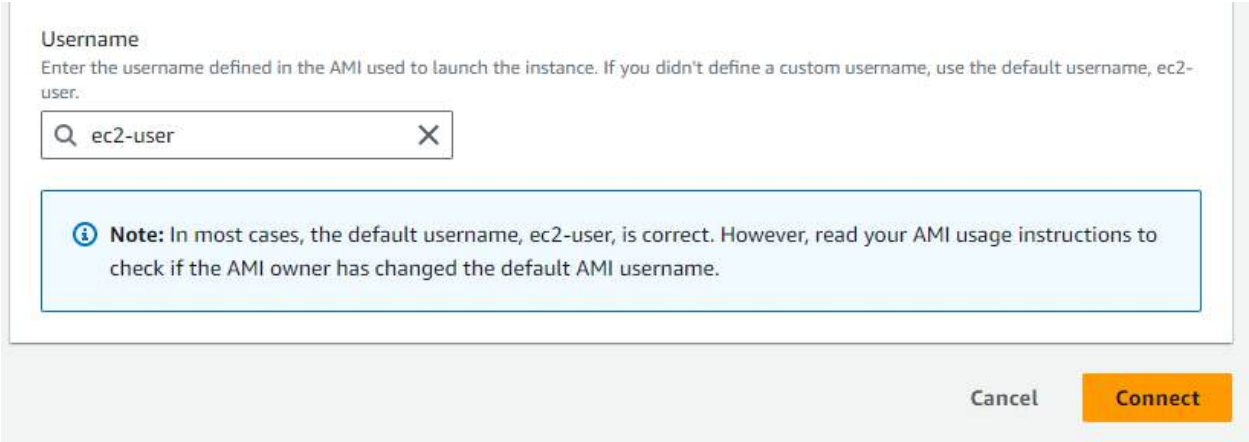


2.12 Now you can see **ec2_neogym** instance click on **INSTANCE ID** and will redirect to a page where you can see **Connect** click on it.



2.13 Under **Connect to instance** four options available click on **EC2 Instance Connect** under Connection type.





2.14 Now go to https://www.free-css.com/free-css-templates/page296/neogym address and right click on the **Download button** and copy the link address.

2.15 Type in the command **“sudo wget (paste link address)”** to pull out the file from the internet and hit enter.

```
  ,     #_
  ~\_  ####_        Amazon Linux 2023
 ~~  \_#####\
 ~~     \###|
 ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
  ~~       V~' '->
   ~~~         /
     ~~._.   _/
        _/ _/
      _/m/'
Last login: Tue Apr 30 05:41:28 2024 from 13.233.177.4
[ec2-user@ip-172-31-40-247 ~]$ sudo wget https://www.free-css.com/assets/files/free-css-templates/download/page296/neogym.zip
--2024-04-30 05:50:38--  https://www.free-css.com/assets/files/free-css-templates/download/page296/neogym.zip
Resolving www.free-css.com (www.free-css.com)... 217.160.0.242, 2001:8d8:100f:f000::28f
Connecting to www.free-css.com (www.free-css.com)|217.160.0.242|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 724954 (708K) [application/zip]
Saving to: ‘neogym.zip’

neogym.zip                          100%[===================================================================>] 707.96K   857KB/s    in 0.8s

2024-04-30 05:50:40 (857 KB/s) - ‘neogym.zip’ saved [724954/724954]
```

2.16 If you want you verify that the pasted link address is pulled out from the internet then you use command “**ls -l”**

```
[ec2-user@ip-172-31-40-247 ~]$ ls -l
total 708
-rw-r--r--. 1 root root 724954 Aug 20  2021 neogym.zip
[ec2-user@ip-172-31-40-247 ~]$
```

2.17 To unzip the file type command **“unzip filename.zip”** and then list the files by **“ls -l**”.

```
[ec2-user@ip-172-31-40-247 ~]$ unzip neogym.zip
Archive:  neogym.zip
   creating: neogym-html/
  inflating: neogym-html/contact.html
   creating: neogym-html/css/
  inflating: neogym-html/css/bootstrap.css
  inflating: neogym-html/css/responsive.css
  inflating: neogym-html/css/style.css
  inflating: neogym-html/css/style.css.map
  inflating: neogym-html/css/style.scss
   creating: neogym-html/images/
  inflating: neogym-html/images/contact-img.jpg
  inflating: neogym-html/images/envelope-white.png
  inflating: neogym-html/images/facebook-logo.png
  inflating: neogym-html/images/healthy-bg.jpg
  inflating: neogym-html/images/hero-bg.jpg
  inflating: neogym-html/images/instagram-logo.png
  inflating: neogym-html/images/location-white.png
  inflating: neogym-html/images/menu.png
  inflating: neogym-html/images/search-icon.png
```

```
[ec2-user@ip-172-31-40-247 ~]$ ls -l
total 708
drwxr-xr-x. 5 ec2-user ec2-user    115 Sep 15  2020 neogym-html
-rw-r--r--. 1 root     root     724954 Aug 20  2021 neogym.zip
[ec2-user@ip-172-31-40-247 ~]$
```

2.18 Next, now **change the directory** to the project folder an install **apache server** by typing command "**sudo yum install httpd -y"**

```
[ec2-user@ip-172-31-40-247 ~]$ cd neogym-html
[ec2-user@ip-172-31-40-247 neogym-html]$
```

2.19 To view the content in that file use command **"ls -l"**

```
[ec2-user@ip-172-31-40-247 neogym-html]$ ls -l
total 60
-rw-r--r--. 1 ec2-user ec2-user  5649 Sep 15  2020 contact.html
drwxr-xr-x. 2 ec2-user ec2-user   105 Sep 15  2020 css
drwxr-xr-x. 2 ec2-user ec2-user 16384 Sep 15  2020 images
-rw-r--r--. 1 ec2-user ec2-user 17875 Sep 15  2020 index.html
drwxr-xr-x. 2 ec2-user ec2-user    53 Sep 15  2020 js
-rw-r--r--. 1 ec2-user ec2-user  6842 Sep 15  2020 trainer.html
-rw-r--r--. 1 ec2-user ec2-user  6738 Sep 15  2020 why.html
[ec2-user@ip-172-31-40-247 neogym-html]$
```

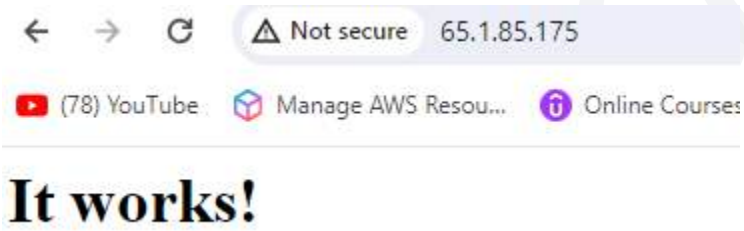
2.20 Now to install **apache server** use the command "**sudo yum install httpd -y"**

```
[ec2-user@ip-172-31-40-247 neogym-html]$ sudo yum install httpd -y
Last metadata expiration check: 0:20:37 ago on Tue Apr 30 05:41:27 2024.
Dependencies resolved.
================================================================================
 Package                 Architecture    Version                    Repository    Size
================================================================================
Installing:
 httpd                   x86_64          2.4.59-2.amzn2023          amazonlinux    47 k
Installing dependencies:
 apr                     x86_64          1.7.2-2.amzn2023.0.2       amazonlinux   129 k
 apr-util                x86_64          1.6.3-1.amzn2023.0.1       amazonlinux    98 k
 generic-logos-httpd     noarch          18.0.0-12.amzn2023.0.3     amazonlinux    19 k
 httpd-core              x86_64          2.4.59-2.amzn2023          amazonlinux   1.4 M
 httpd-filesystem        noarch          2.4.59-2.amzn2023          amazonlinux    14 k
 httpd-tools             x86_64          2.4.59-2.amzn2023          amazonlinux    81 k
 libbrotli               x86_64          1.0.9-4.amzn2023.0.2       amazonlinux   315 k
 mailcap                 noarch          2.1.49-3.amzn2023.0.3      amazonlinux    33 k
Installing weak dependencies:
 apr-util-openssl        x86_64          1.6.3-1.amzn2023.0.1       amazonlinux    17 k
 mod_http2               x86_64          2.0.27-1.amzn2023.0.2      amazonlinux   166 k
```

2.21 To run the server use **"sudo systemctl start httpd"**

```
[ec2-user@ip-172-31-40-247 neogym-html]$ sudo systemctl start httpd
[ec2-user@ip-172-31-40-247 neogym-html]$
```

2.22 To verify the web page is accessing to do this copy the webpage URL and then open new TAB and then pasted, now it shows as following



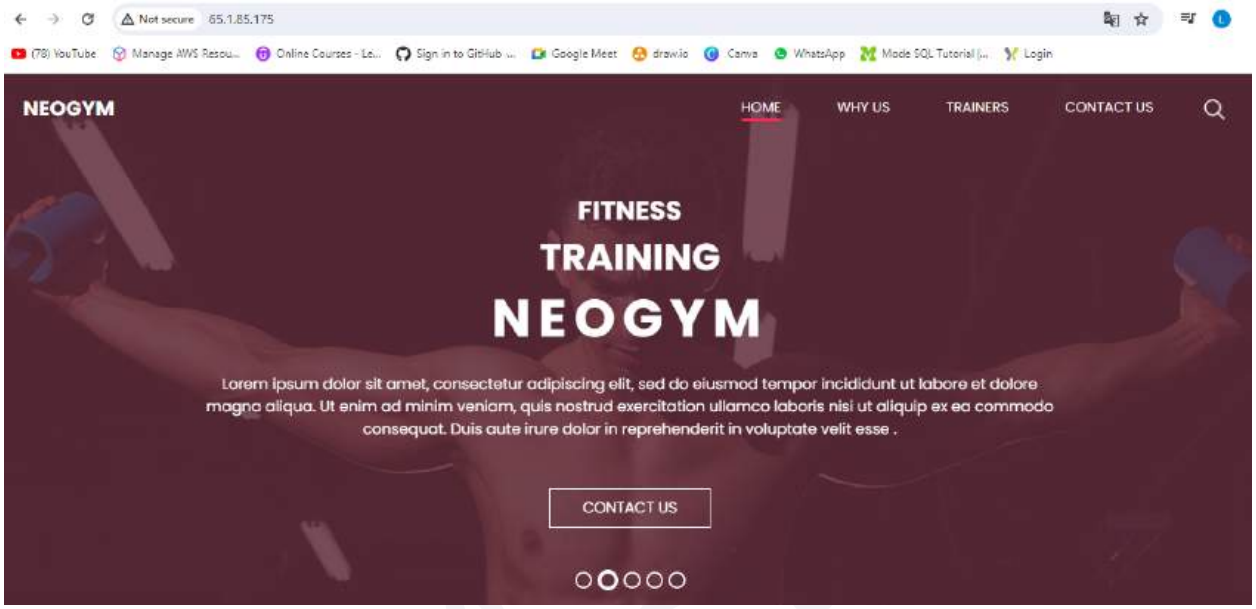
2.23 Now use the command **"sudo mv '.' /var/www/html"** to move the folders to the base directory of the website and then list the files by **"ls -l"**.

```
[ec2-user@ip-172-31-40-247 neogym-html]$ sudo mv *.* /var/www/html/
[ec2-user@ip-172-31-40-247 neogym-html]$ ls -l
total 16
drwxr-xr-x. 2 ec2-user ec2-user   105 Sep 15  2020 css
drwxr-xr-x. 2 ec2-user ec2-user 16384 Sep 15  2020 images
drwxr-xr-x. 2 ec2-user ec2-user    53 Sep 15  2020 js
[ec2-user@ip-172-31-40-247 neogym-html]$
```

2.24 Similarly, now move the other folders by **"sudo mv css/ images/ js/ /var/www/html"**.

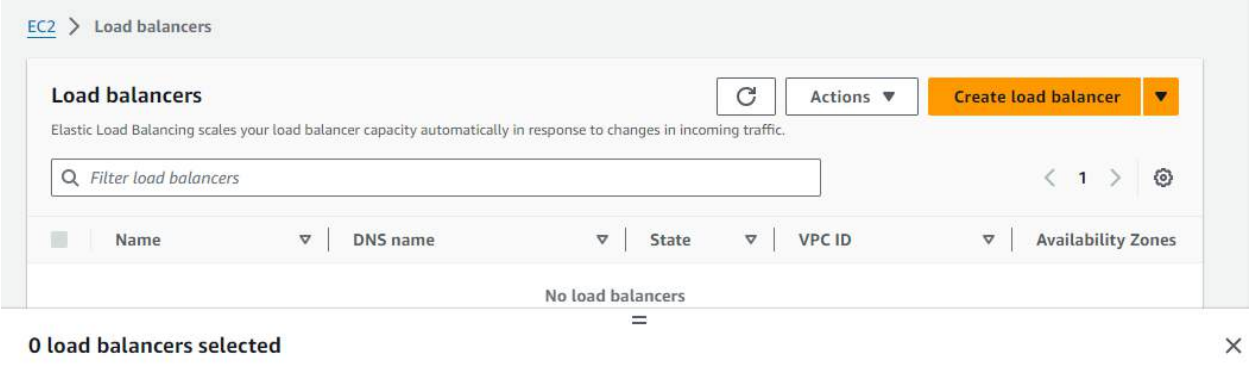
```
[ec2-user@ip-172-31-40-247 neogym-html]$ sudo mv css/ images/ js/ /var/www/html/
[ec2-user@ip-172-31-40-247 neogym-html]$
```

2.25 Successfully the website is hosted with a **Public IP address.**

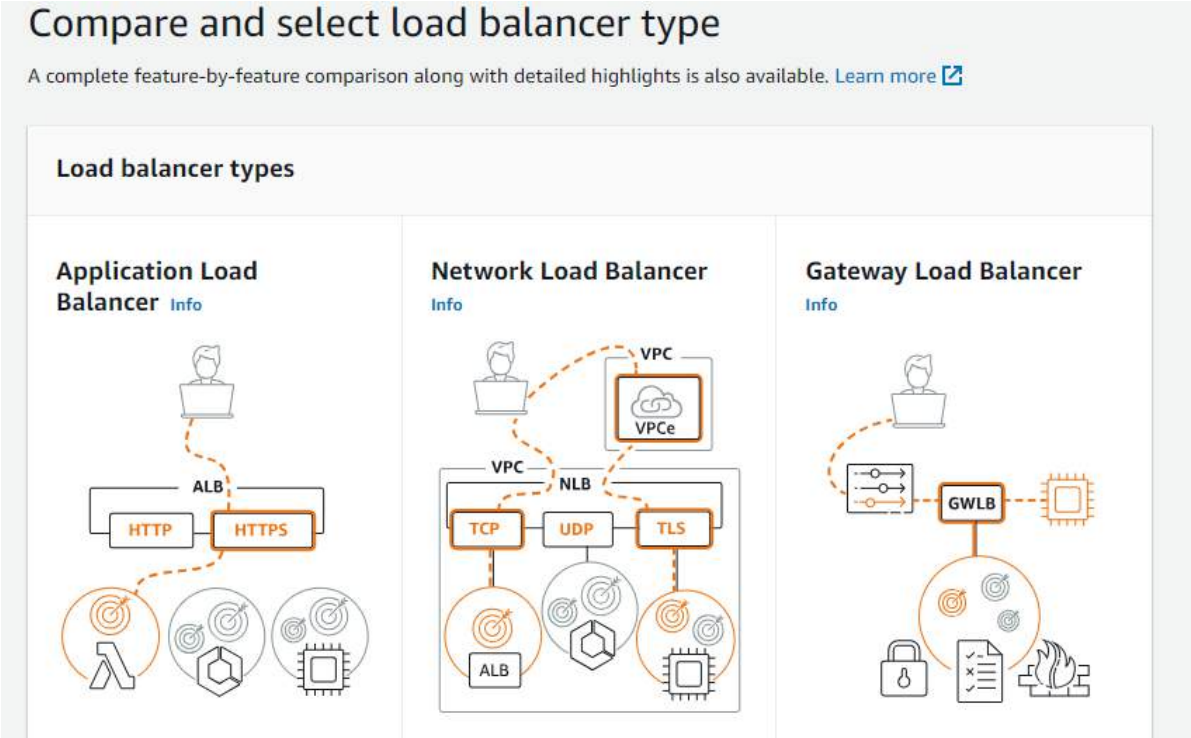


Step 3: Launch ELB to distribute the load

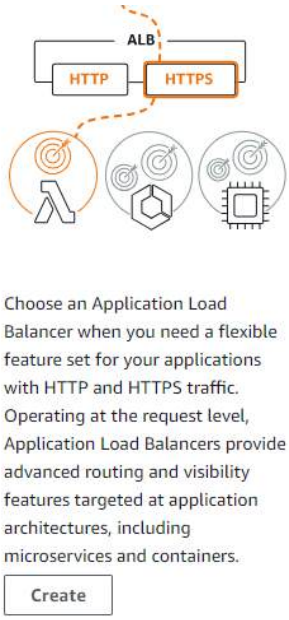
3.1 In the EC2 left pane scroll down to find **Load Balancer** click on it and create load balancer.



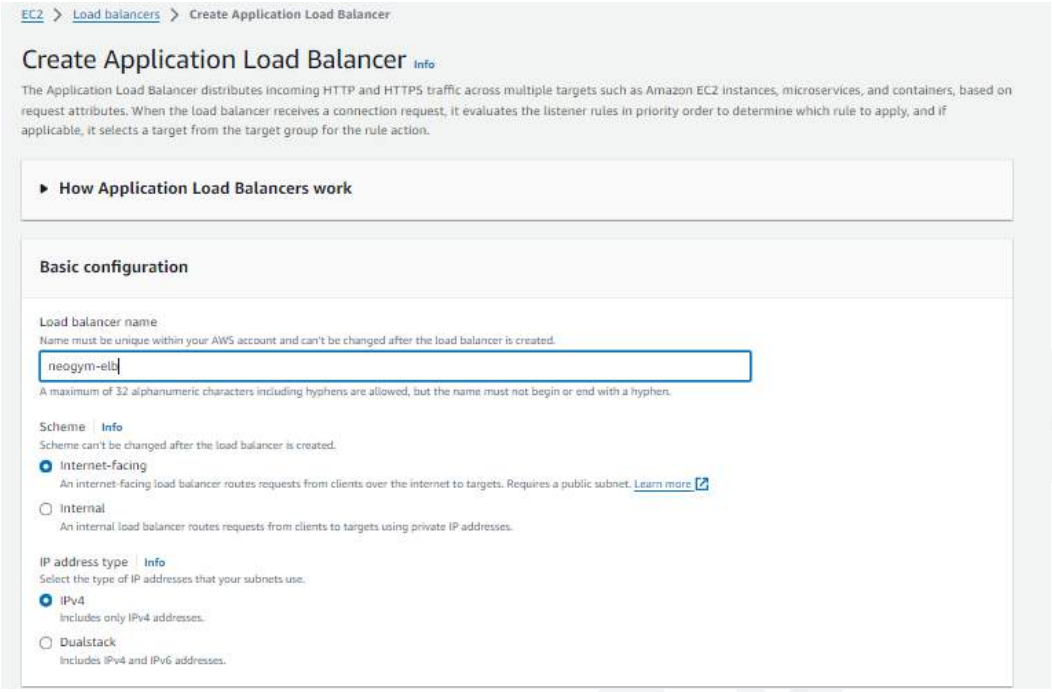
3.2 So here, we will choose **Application Load Balancer** to request HTTP traffic.



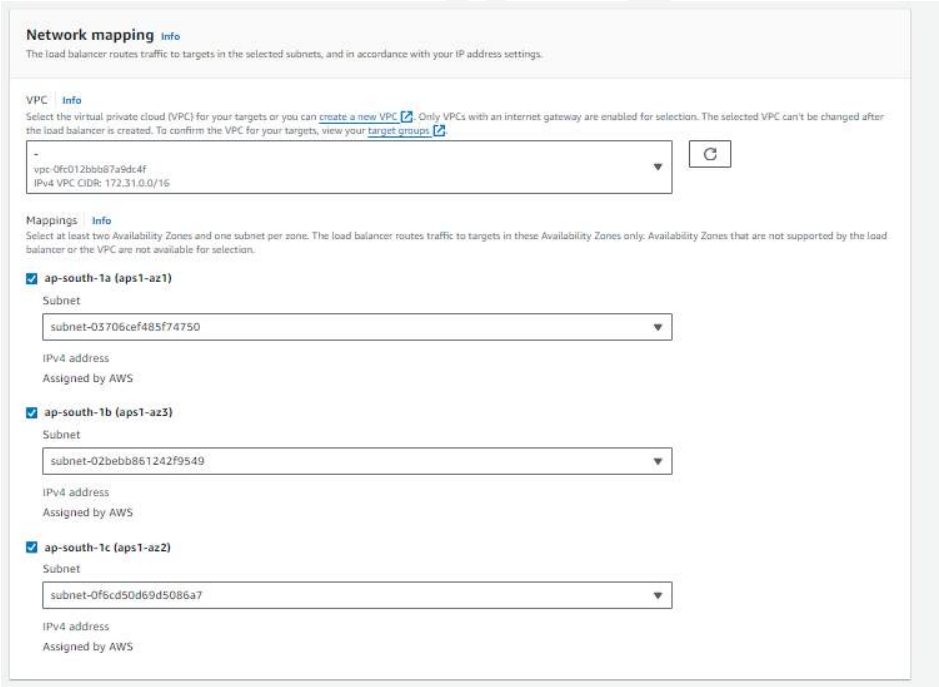
Click on **Create**



3.3 Enter the name **"neogym-elb"** for the application name.



3.4 Add the VPC and the mappings for Availability zones with the existing security group and create a **Target group**.



Security groups Info

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can create a new security group.

Security groups

Select up to 5 security groups

SG-NG
sg-0ac05a51b10901aa5 VPC: vpc-0fc012bbb87a9dc4f

Listeners and routing Info

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener **HTTP:80** Remove

Protocol	Port	Default action Info
HTTP	80 (1-65535)	Forward to *Select a target group*

Create target group

Listener tags - *optional*

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

Add listener tag

You can add up to 50 more tags.

EC2 > Target groups > **Create target group**

Step 1
Specify group details

Step 2
Register targets

Specify group details

Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Basic configuration

Settings in this section can't be changed after the target group is created.

Choose a target type

- (●) Instances
  - Supports load balancing to instances within a specific VPC.
  - Facilitates the use of Amazon EC2 Auto Scaling to manage and scale your EC2 capacity.
- ( ) IP addresses
  - Supports load balancing to VPC and on-premises resources.
  - Facilitates routing to multiple IP addresses and network interfaces on the same instance.
  - Offers flexibility with microservice based architectures, simplifying inter-application communication.
  - Supports IPv6 targets, enabling end-to-end IPv6 communication, and IPv4-to-IPv6 NAT.
- ( ) Lambda function
  - Facilitates routing to a single Lambda function.
  - Accessible to Application Load Balancers only.
- ( ) Application Load Balancer
  - Offers the flexibility for a Network Load Balancer to accept and route TCP requests within a specific VPC.

Target group name

neogym-tg

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Protocol : Port

Choose a protocol for your target group that corresponds to the Load Balancer type that will route traffic to it. Some protocols now include anomaly detection for the targets and you can set mitigation options once your target group is created. This choice cannot be changed after creation

HTTP | 80

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

- (●) IPv4
  Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.
- ( ) IPv6
  Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). Learn more

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

-
vpc-0fc012bbb87a9dc4f
IPv4 VPC CIDR: 172.31.0.0/16

Protocol version

3.5 Click on **Next** to create Load Balancer

Attributes

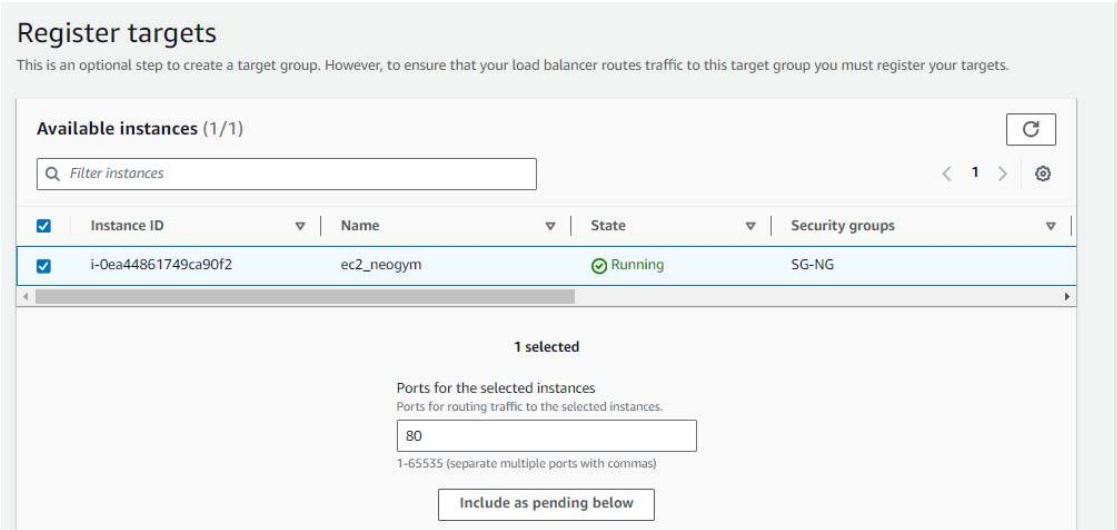
Certain default attributes will be applied to your target group. You can view and edit them after creating the target group.

▸ **Tags - *optional***

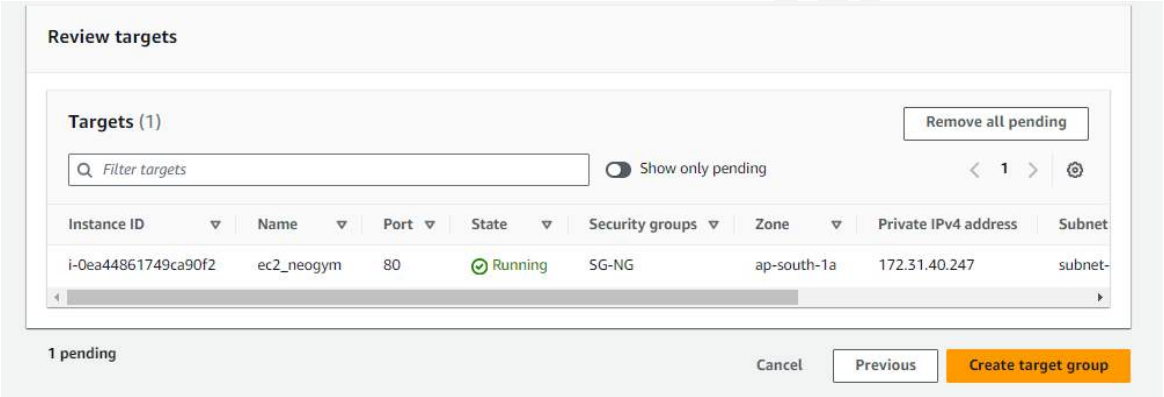
Consider adding tags to your target group. Tags enable you to categorize your AWS resources so you can more easily manage them.

Cancel | Next

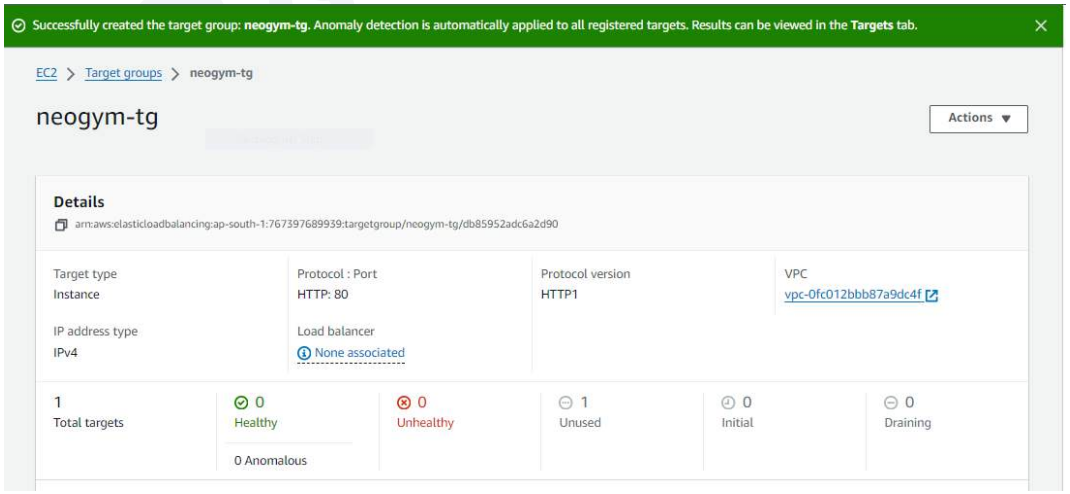
3.6 click on **Include as pending below** to add target group



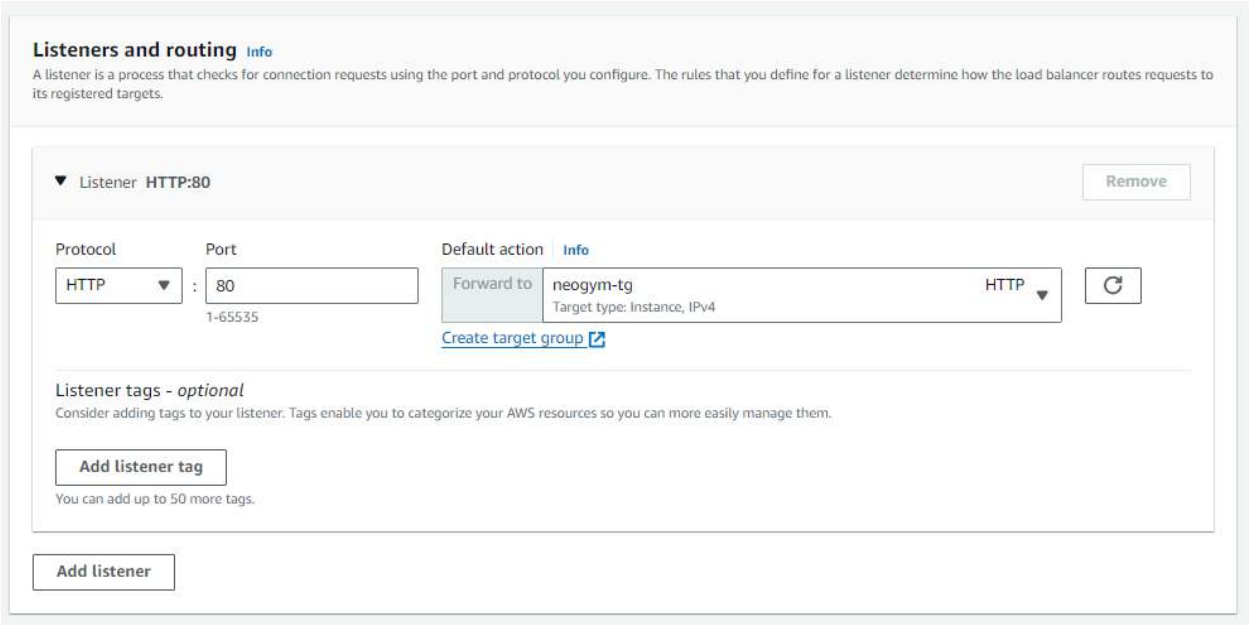
Now Target group have **Registered**

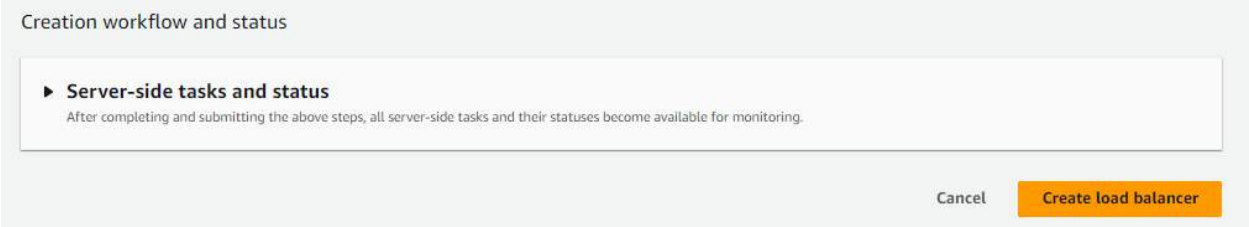


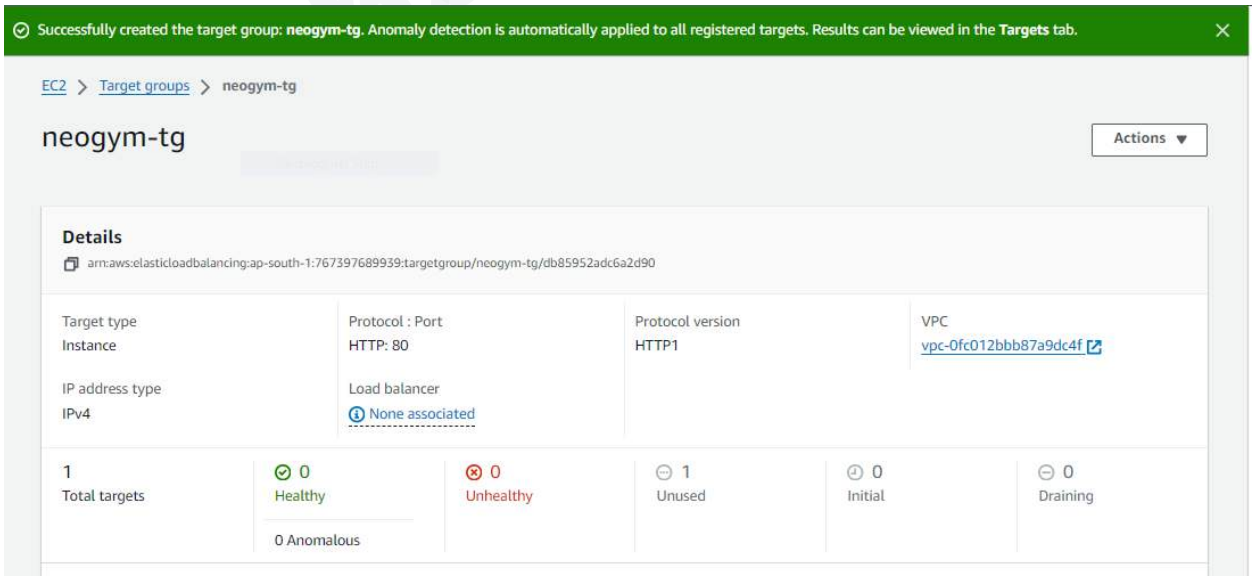
3.7 Click on **Create Target Group**.



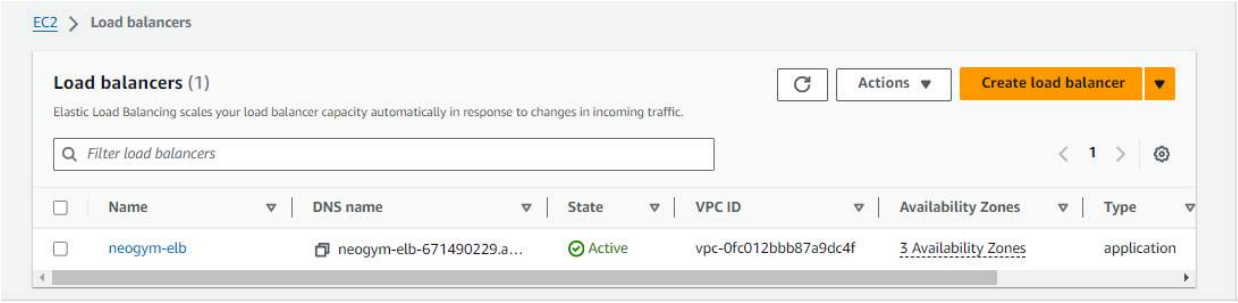
3.8 Now associate the build **Target group** in Application load balancer and click **Create Load balancer**.





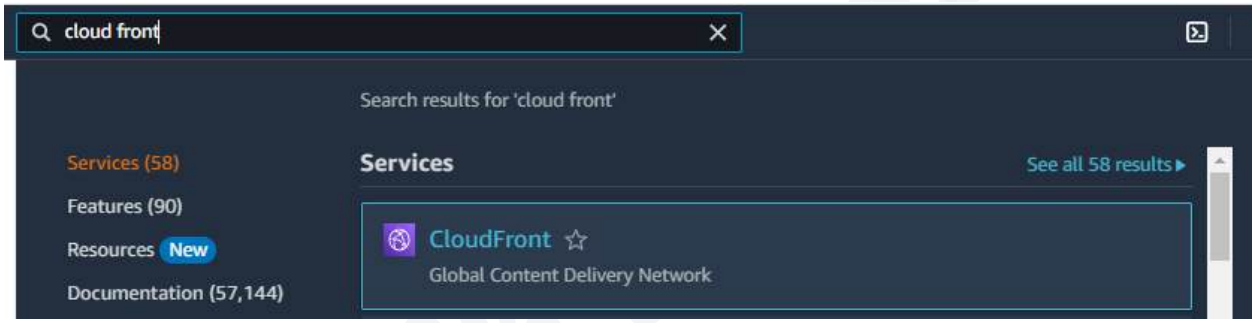


3.9 Now copy the **DNS name** and paste in the New tab to check the load balancing of two instances.

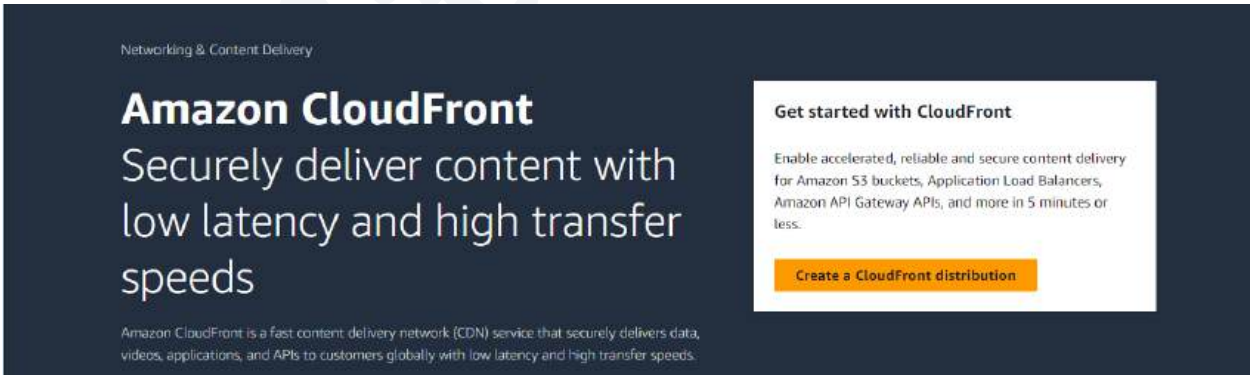


Step 4: Launch CloudFront for distribution of web content

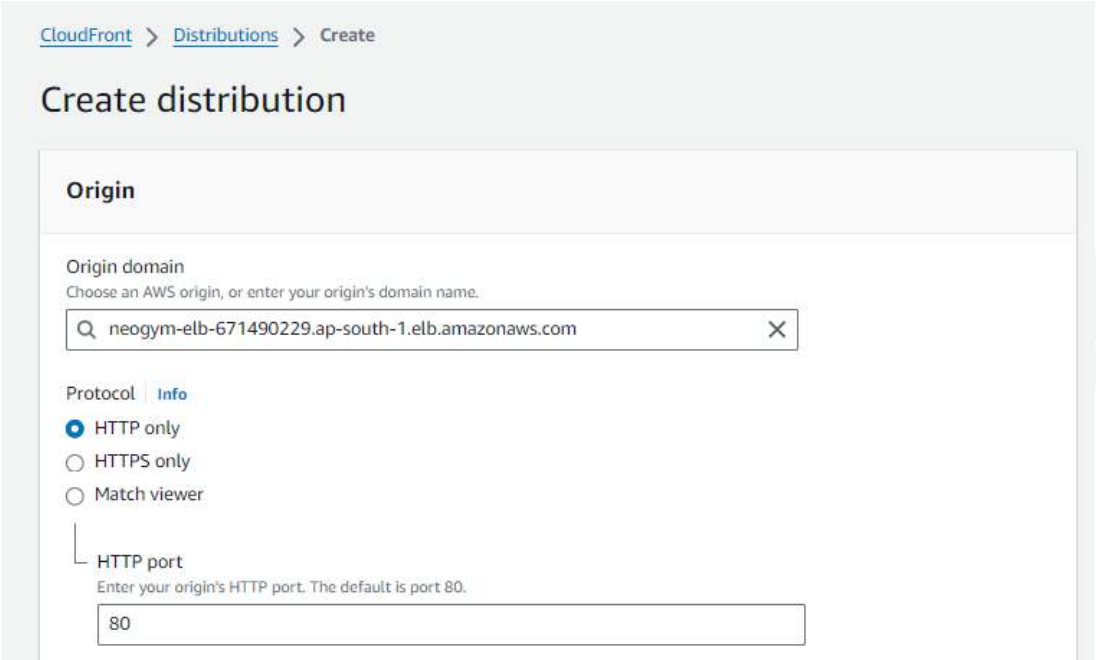
4.1 In Dashboard, search for **Cloudfront service**, click CloudFront to open it.



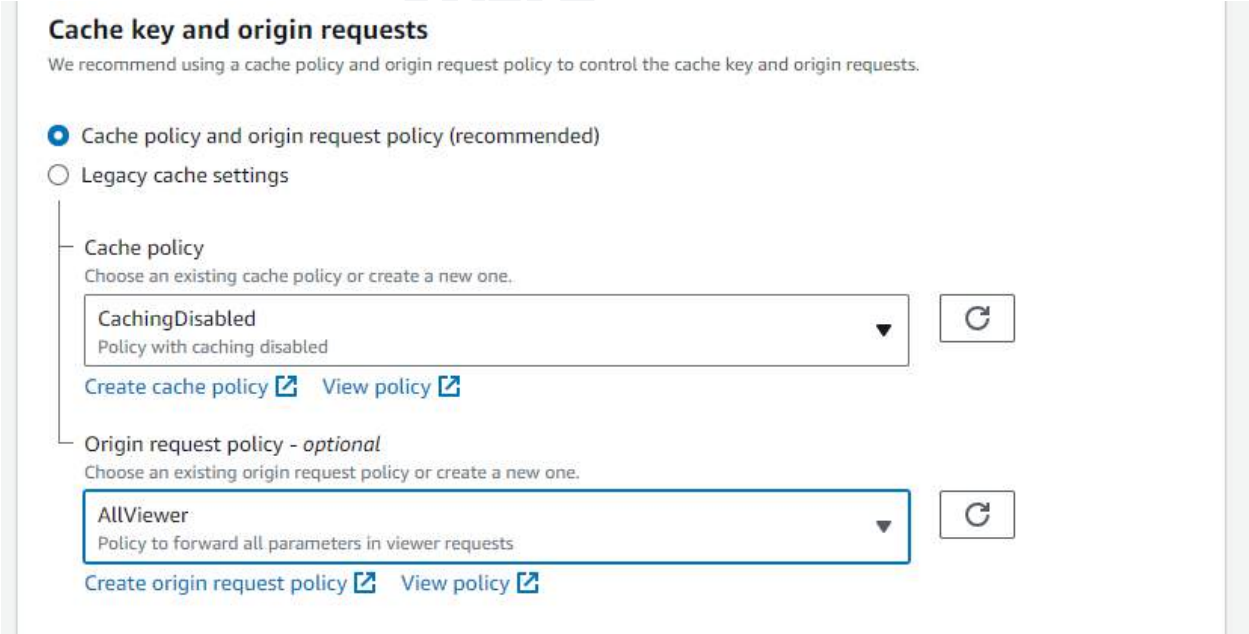
4.2 Click **create a cloudfront distribution** to configure.



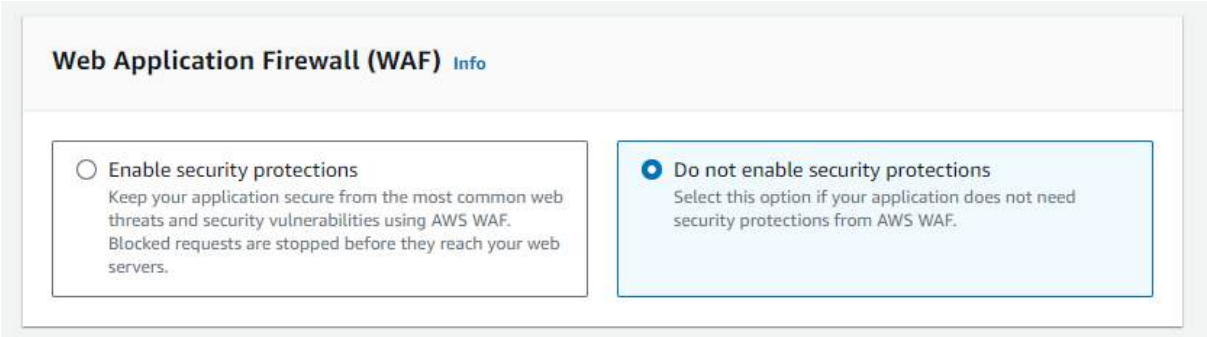
4.3 Under the **origin domain** select the load balancer which was created prior.



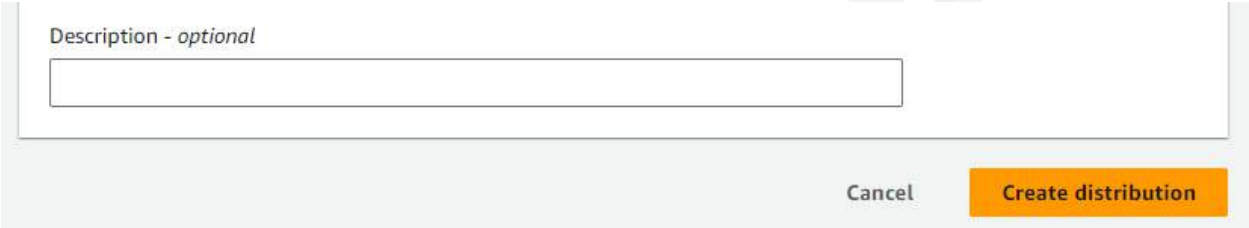
4.4 Now, scroll down further to **Cache key and origin requests** here CachePolicy make it as **CachingDisabled** and Origin request policy as **AllViewer**.



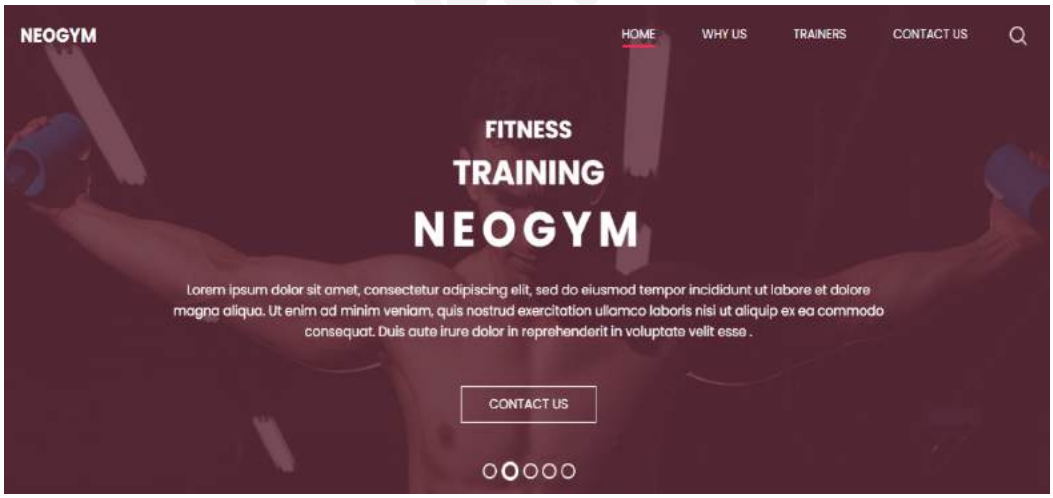
4.5 Disable the WAF by selecting **Do not enable security protections** and rest leave as default and click **Create distribution.**



4.6 click on **Create distribution** so that CloudFront Distribution will be creating and Successfully **created CloudFront new distribution**.



4.7 Now copy the **Distribution domain name** and paste it in a new tab to run the frontend website.

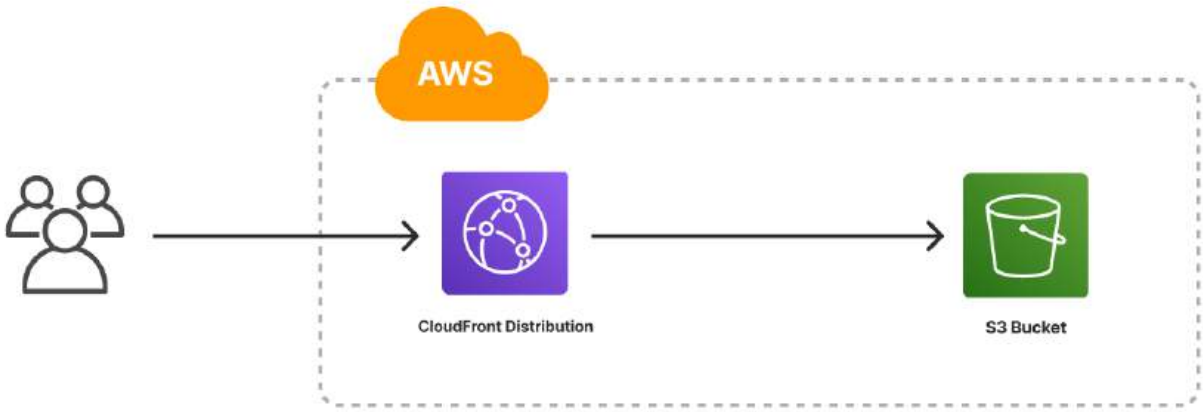


Result 1:

Task on to launch a website without using S3 via CloudFront, EC2 and ELB services successfully accomplished.

Case 2: Using **Amazon S3** for storing the files which are used to host a static website(Amazon S3 is used to store the static files (HTML, CSS, JavaScript, images, etc.) and **Amazon CloudFront** for content distribution.

Architecture:



Data Flow:

a. The user (such as a web browser) sends a request to access the website by entering the website's domain name into the browser's address bar (URL).

b. The request is first routed to the nearest CloudFront edge location. CloudFront acts as a Content Delivery Network (CDN), caching content and serving it from edge locations to reduce latency and improve performance.

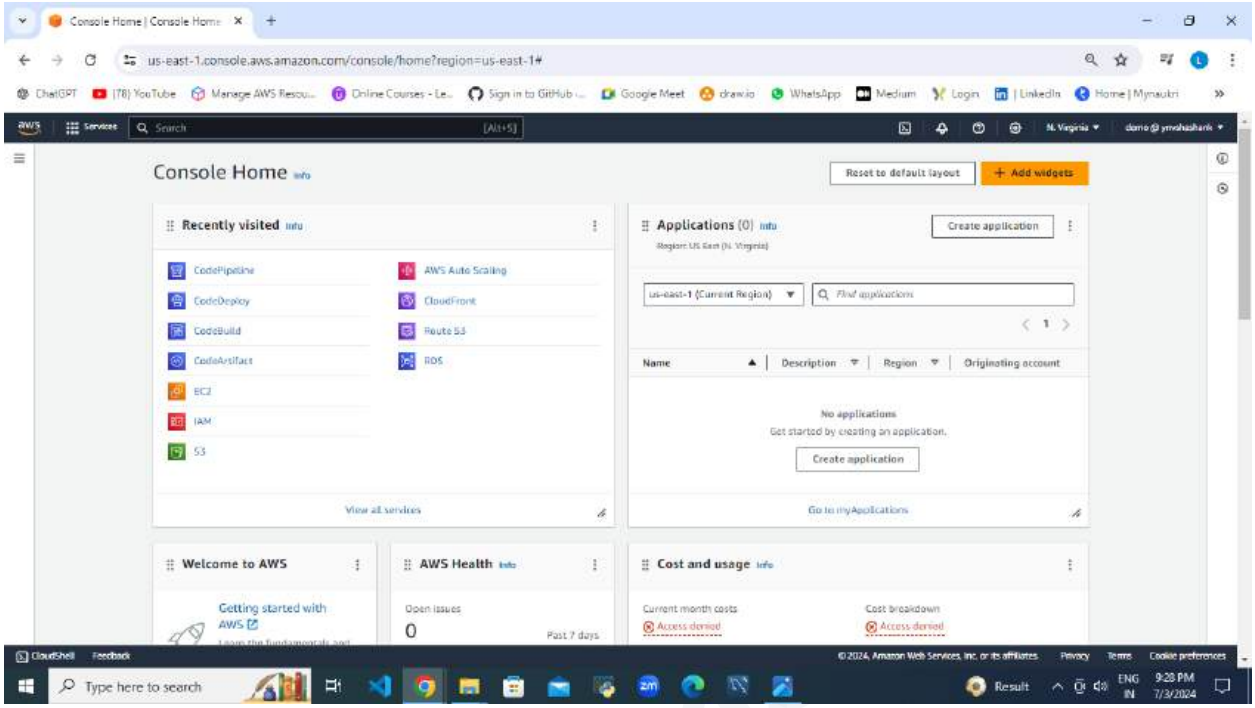
c. To access the content in the S3 Bucket by CloudFront, first it verifies the Bucket Policy is attached to it, like is the necessary permissions are given to CloudFront by S3 Bucket

d. If the S3 Bucket Policy Is correctly attached, then when the CloudFront Distribution domain name is entered on the web browser's URL, then the users can access the static website.

Steps:

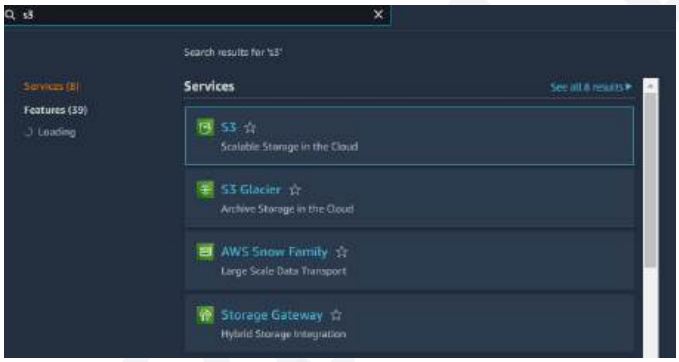
Step 1: Login into AWS Console

1.1 Create an account and login into it by providing the credentials.



Step 2: Create an S3 Bucket in Console

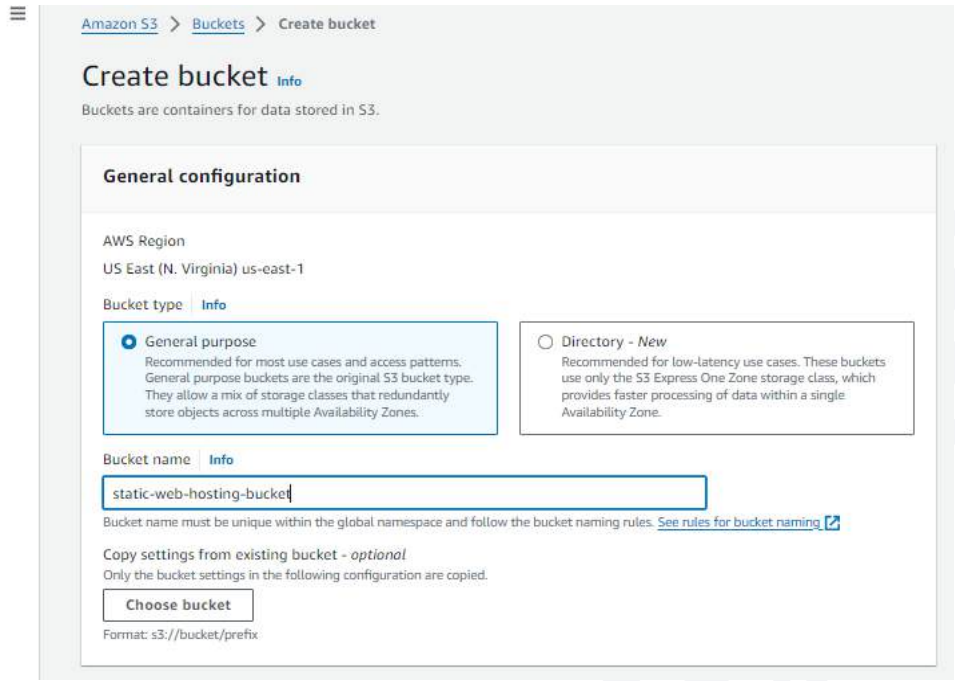
2.1 In Dashboard, search for **Amazon S3 service,** click on **S3** to open it.



2.2 To create an S3 Bucket, Click on **Create Bucket.**

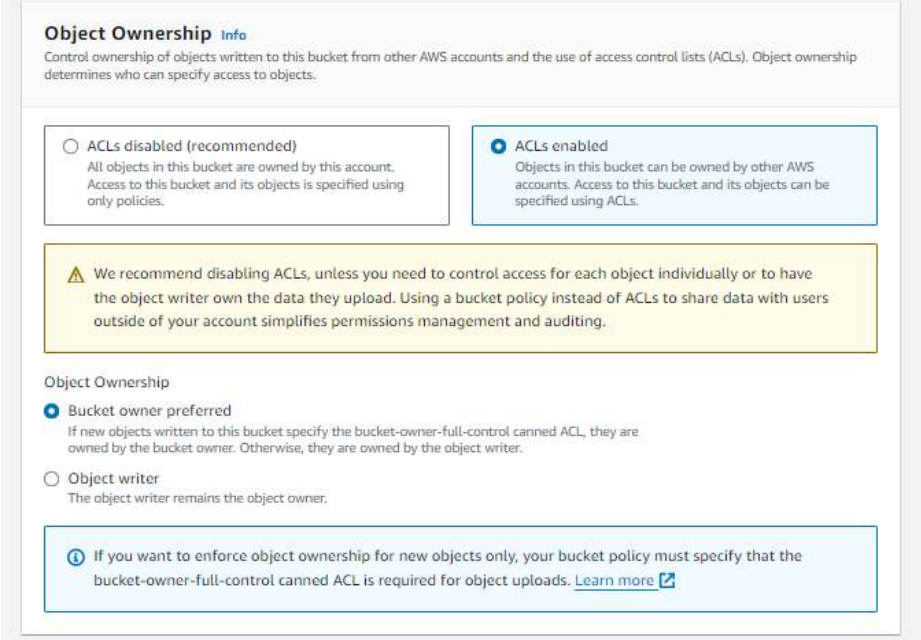
2.3 In **General Configuration** of Bucket, select **Bucket type,** as **General Purpose** and enter the **Bucket name**

NOTE: Bucket name must be unique

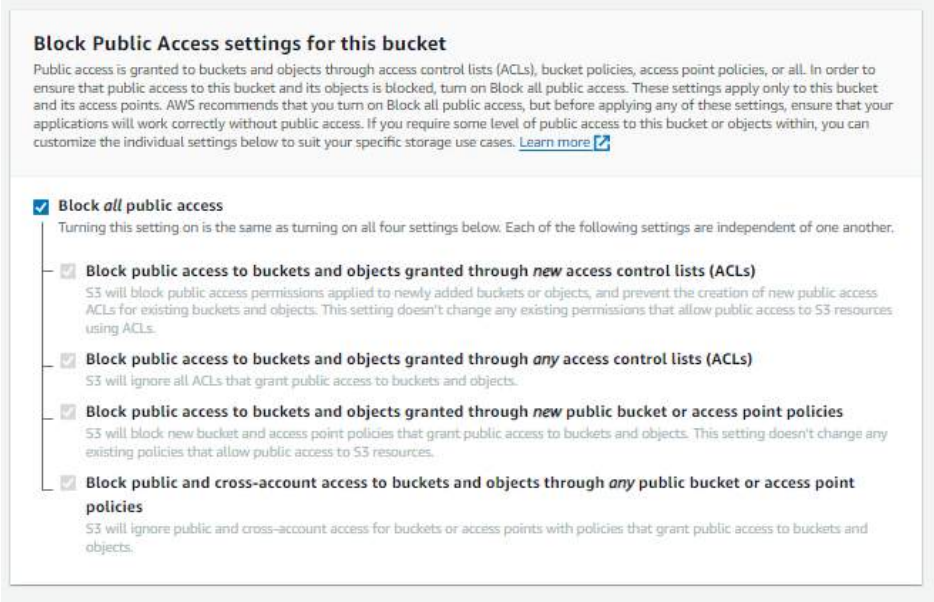


2.4 In Object Ownership, click on ACLs enabled

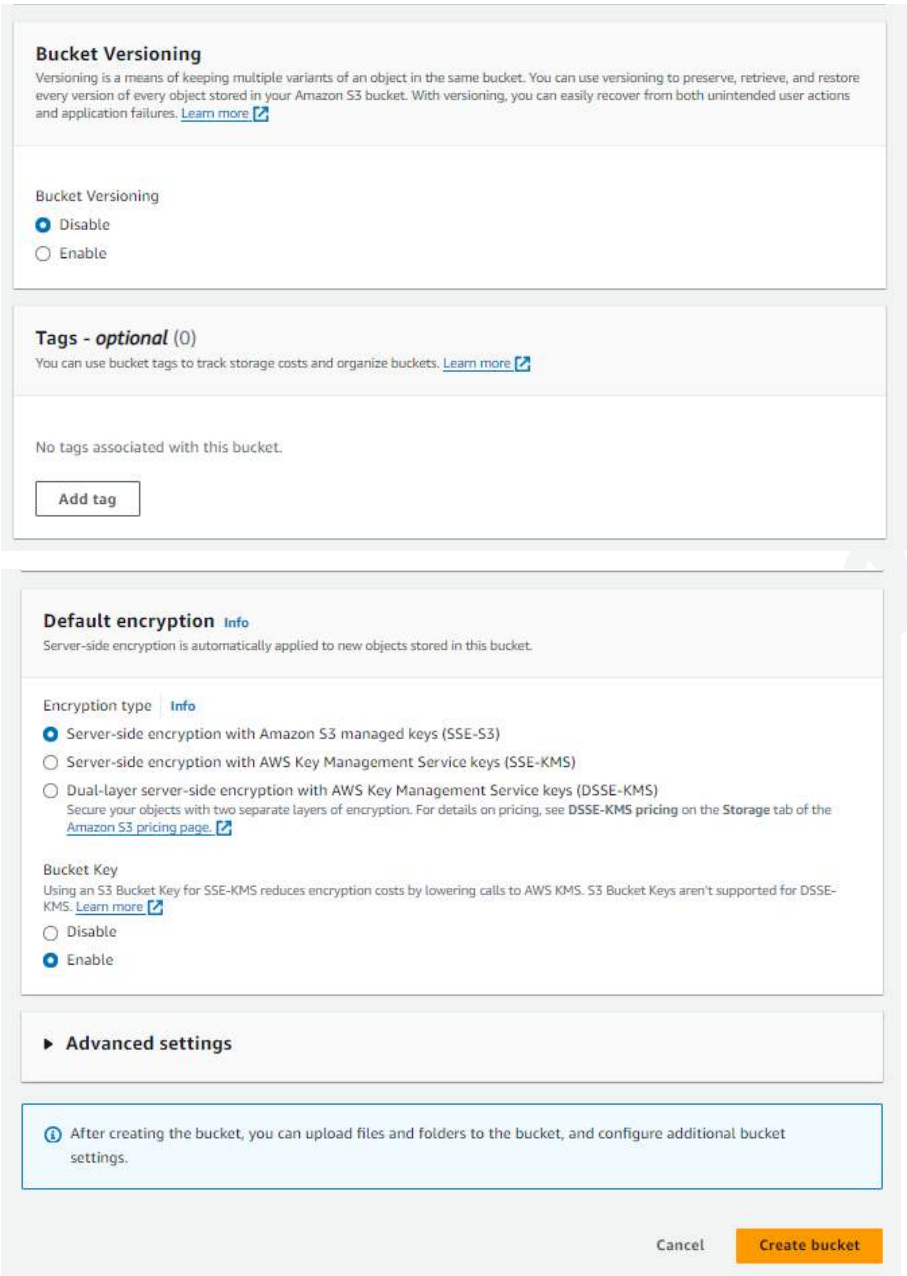
Amazon S3 access control lists (ACLs) enables you to manage access to S3 buckets and objects. Every S3 bucket and object has an ACL attached to it as a sub-resource. The ACLs define which AWS accounts or groups are granted access along with the type of access.



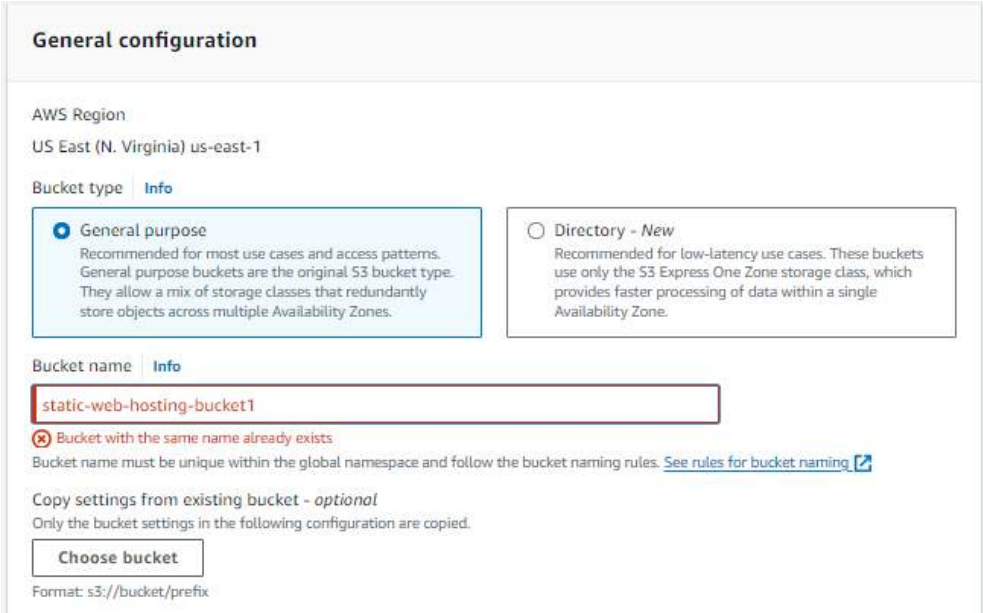
2.5 Now Click on **Block all public access**, which makes only the owner of the bucket can access the objects in the bucket.



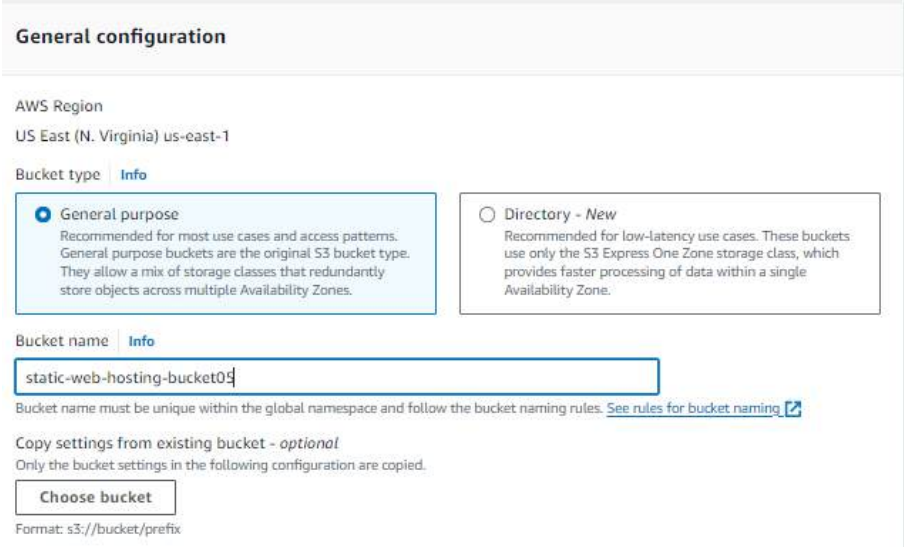
2.6 Keep the remaining settings as it is and then Click on Create Bucket

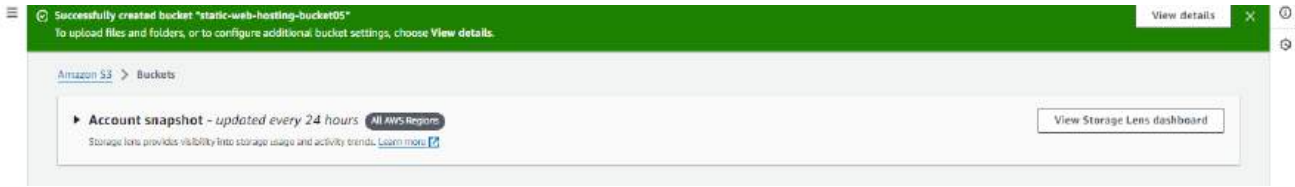


Troubleshooting:

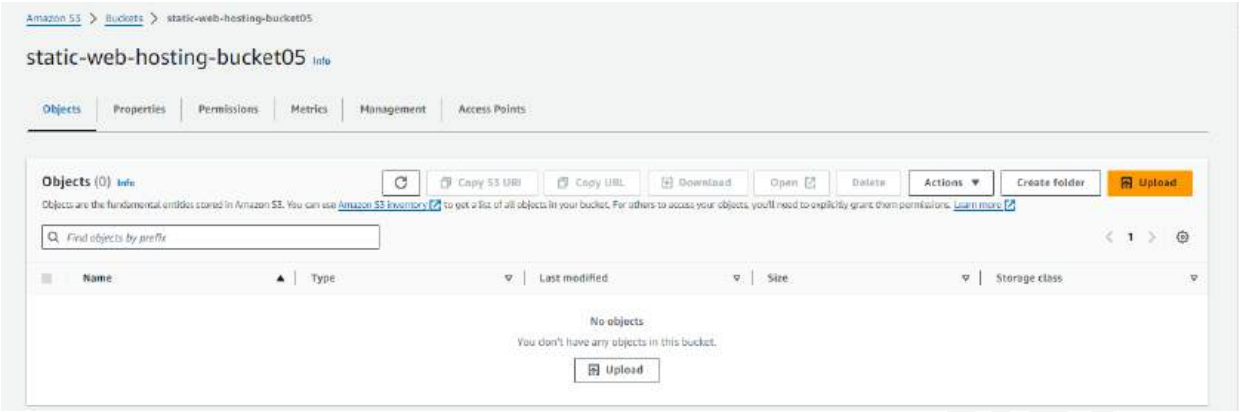


➢ As we know Bucket name must be **Unique,** because S3 Buckets Are Globally Accessed.

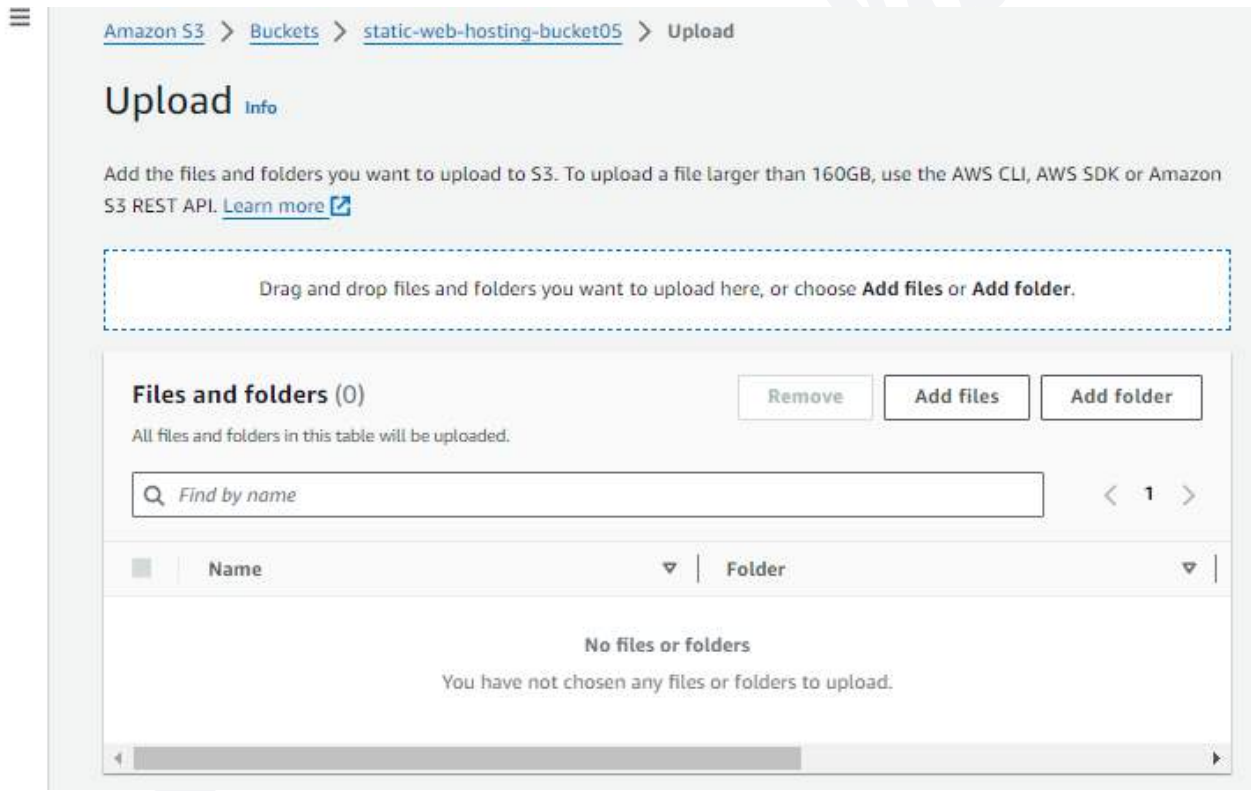


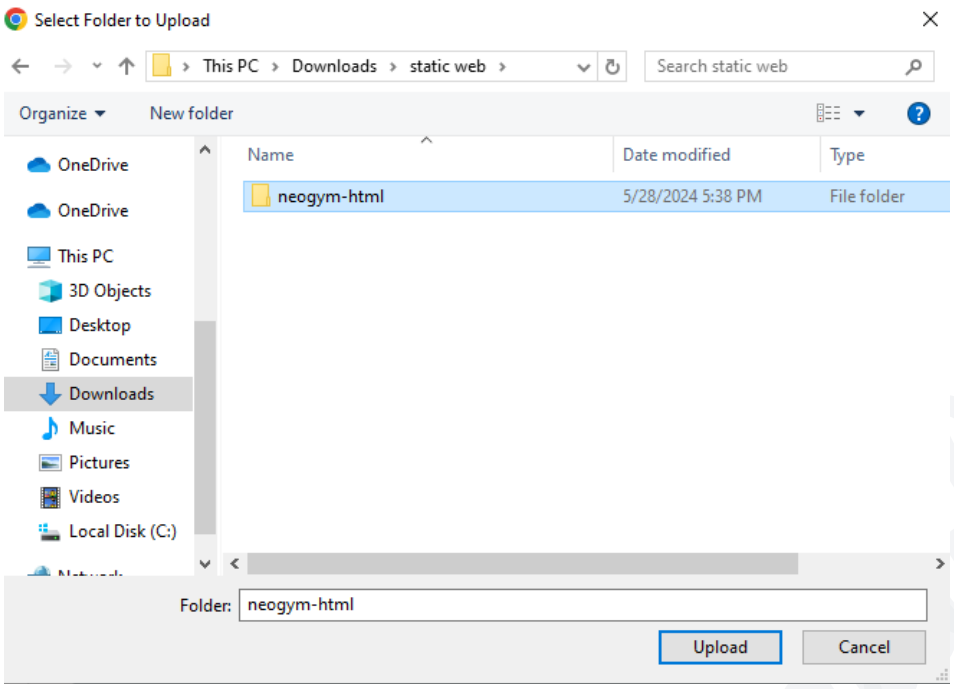


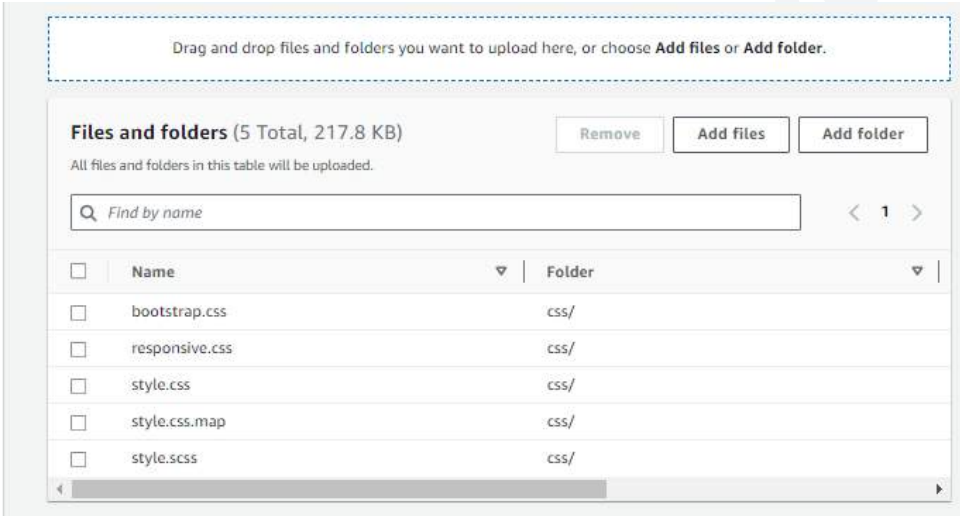
2.7 To Upload files/objects in the Bucket, Click on **Upload**.

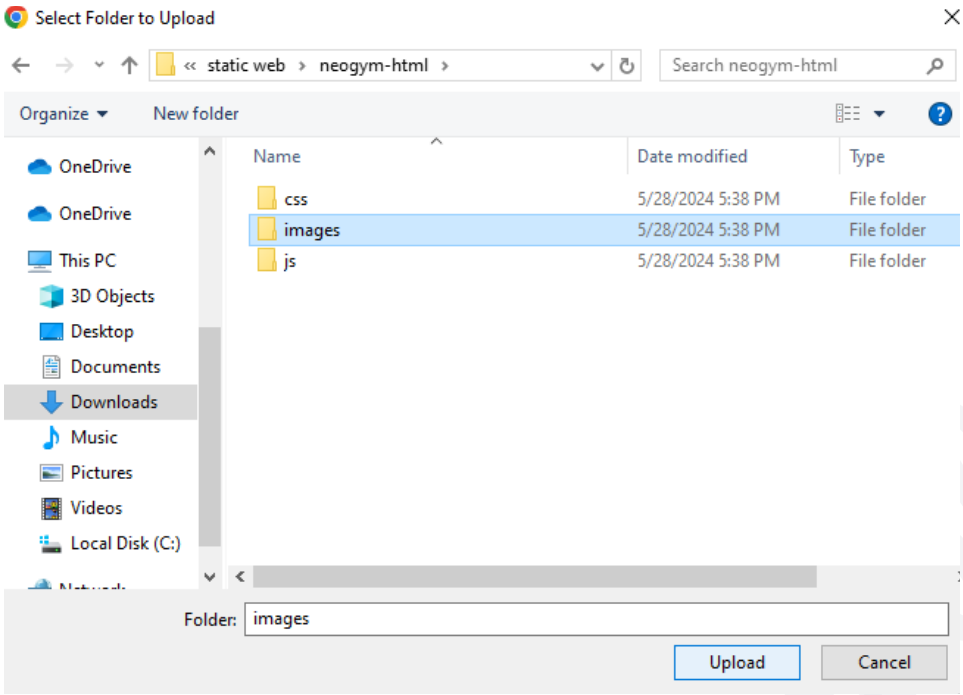


2.8 Now Upload the objects/files into the bucket




Select Folder to Upload
This PC › Downloads › static web ›
Search static web
Organize
New folder
OneDrive
OneDrive
This PC
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
Local Disk (C:)
Name
Date modified
Type
neogym-html
5/28/2024 5:38 PM
File folder
Folder:
neogym-html
Upload
Cancel


Drag and drop files and folders you want to upload here, or choose Add files or Add folder.
Files and folders (5 Total, 217.8 KB)
Remove
Add files
Add folder
All files and folders in this table will be uploaded.
Find by name
1
Name
Folder
bootstrap.css
css/
responsive.css
css/
style.css
css/
style.css.map
css/
style.scss
css/


Select Folder to Upload
static web › neogym-html ›
Search neogym-html
Organize
New folder
OneDrive
OneDrive
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3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
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images
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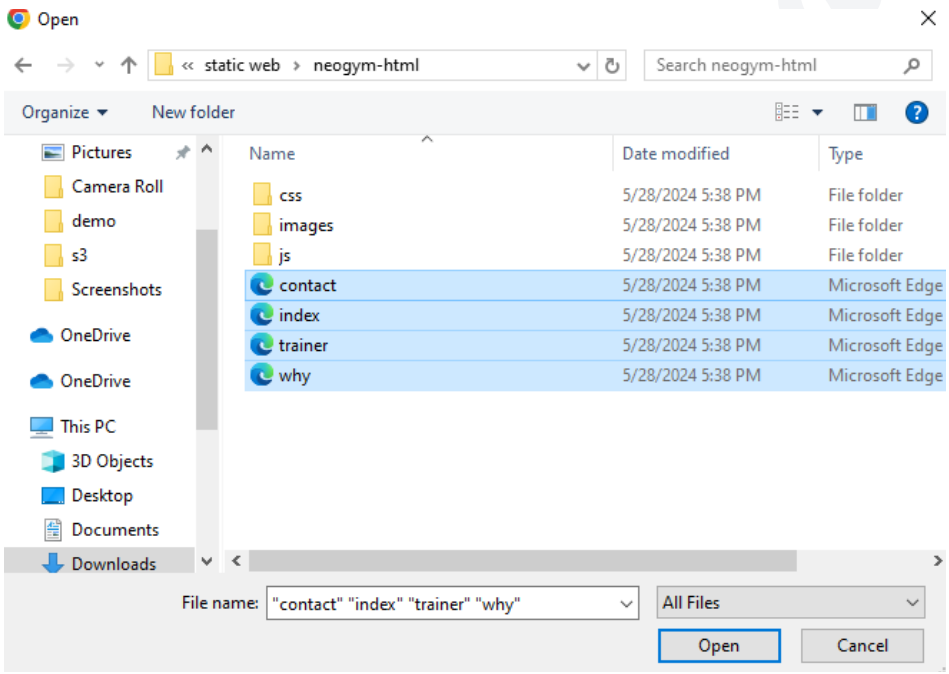
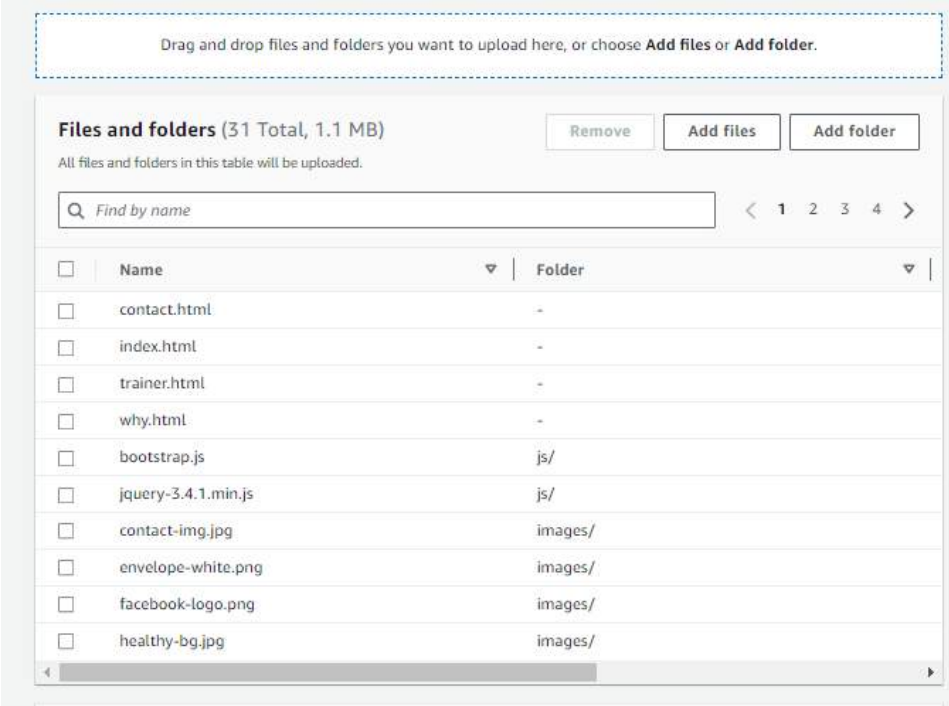
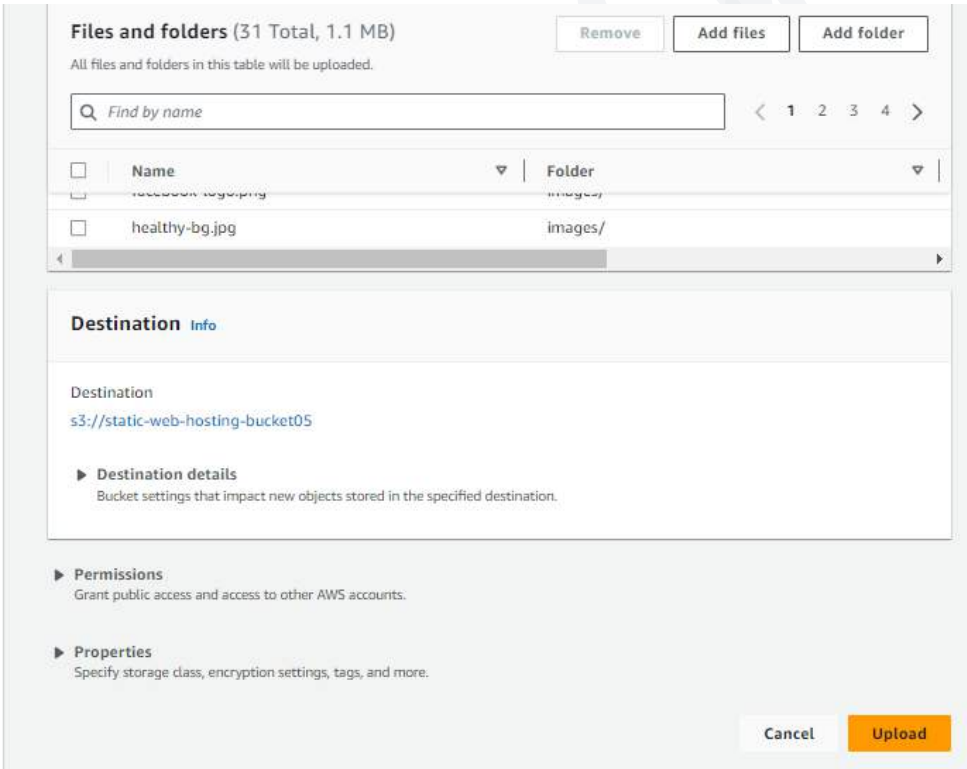
Cancel

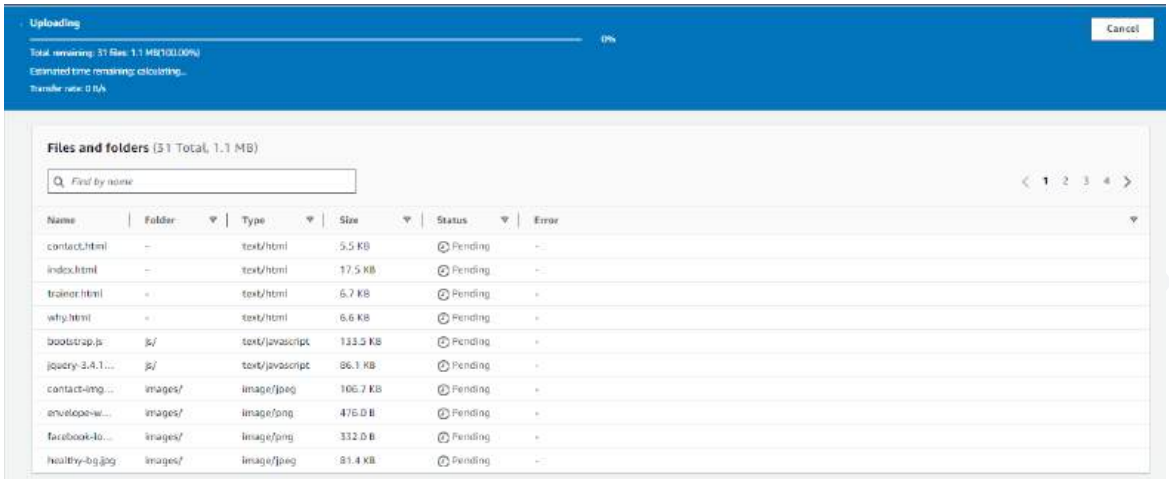

Open
static web › neogym-html
Search neogym-html
Organize
New folder
Pictures
Camera Roll
demo
s3
Screenshots
OneDrive
OneDrive
This PC
3D Objects
Desktop
Documents
Downloads
Name
Date modified
Type
css
5/28/2024 5:38 PM
File folder
images
5/28/2024 5:38 PM
File folder
js
5/28/2024 5:38 PM
File folder
contact
5/28/2024 5:38 PM
Microsoft Edge
index
5/28/2024 5:38 PM
Microsoft Edge
trainer
5/28/2024 5:38 PM
Microsoft Edge
why
5/28/2024 5:38 PM
Microsoft Edge
File name:
"contact" "index" "trainer" "why"
All Files
Open
Cancel



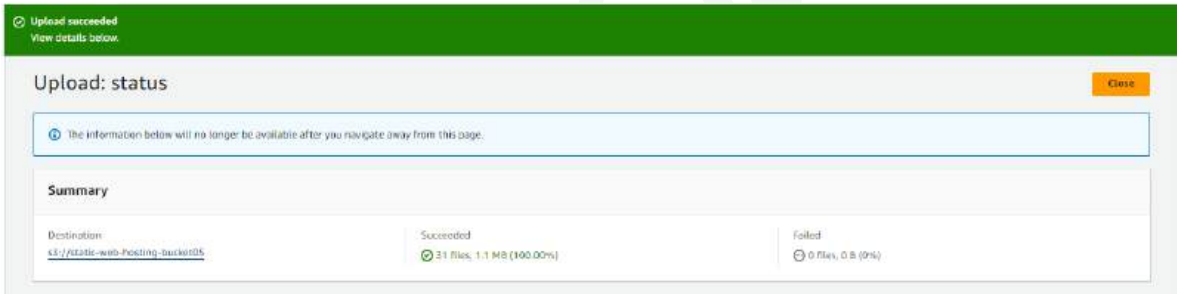
2.9 Now Click on **Upload**.

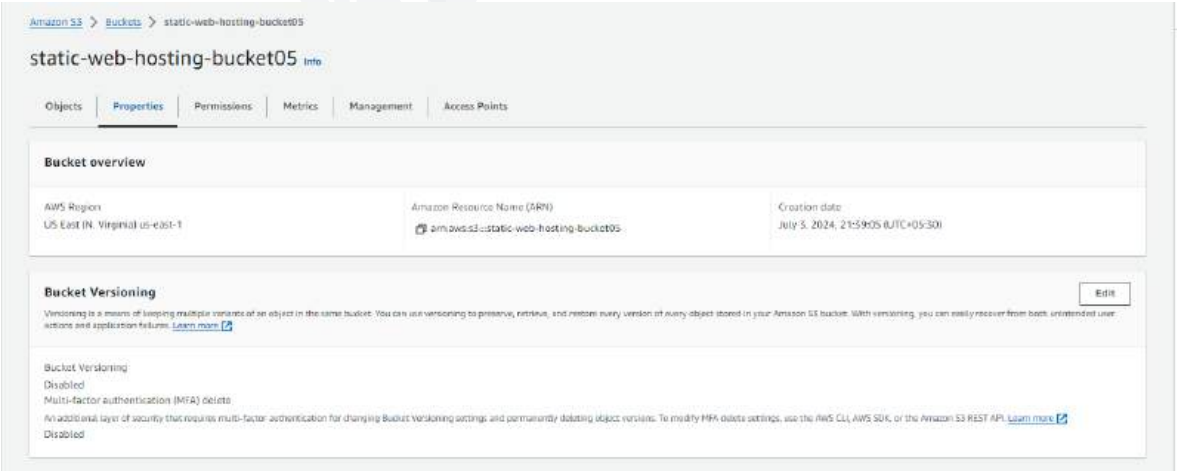


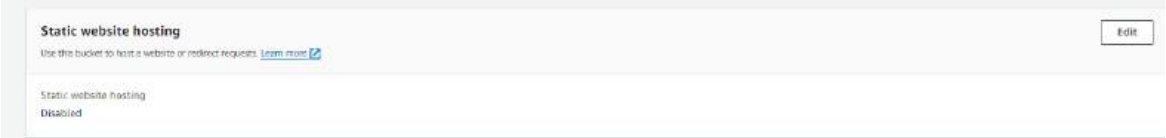
Check the **status** of Objects **Uploading**.



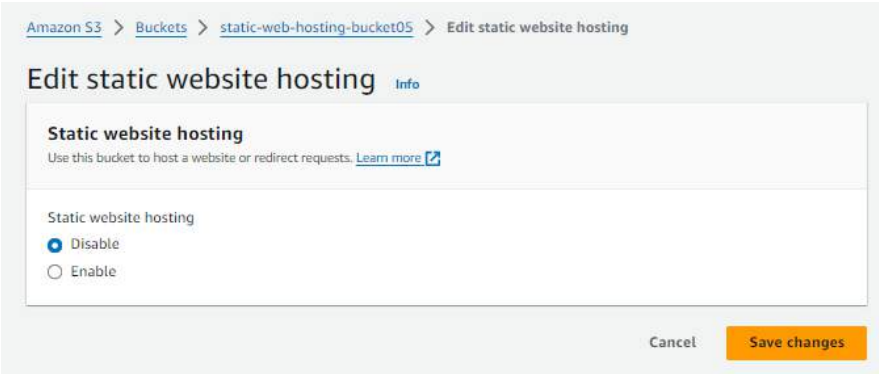
2.10 Click on **Destination** of the Bucket, then Click on **Properties**, then go to **Static website hosting.**

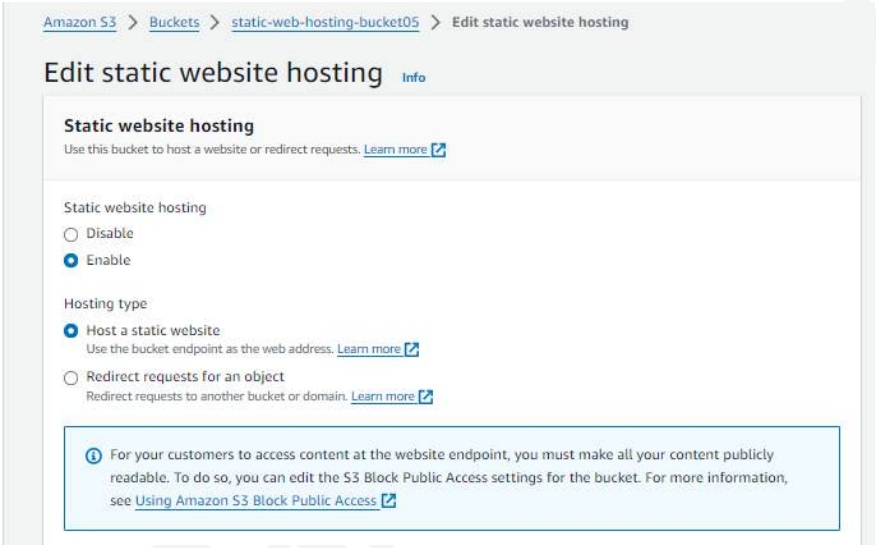


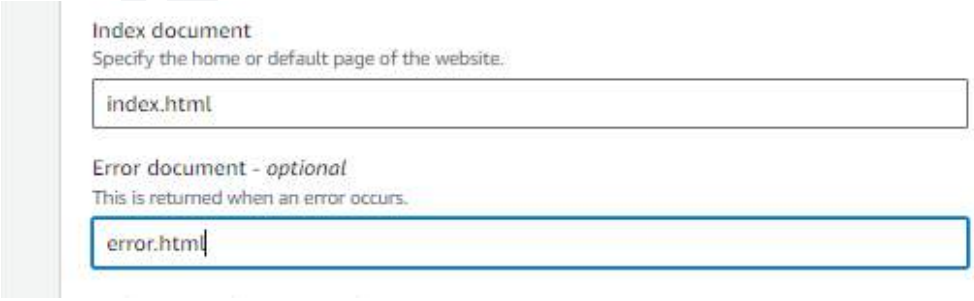


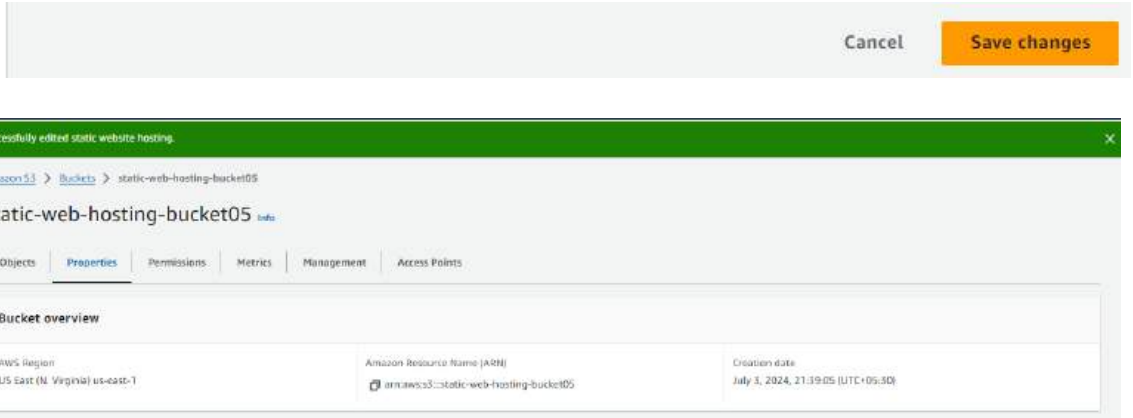


2.11 Edit Static website hosting from **Disable** to **Enable.**

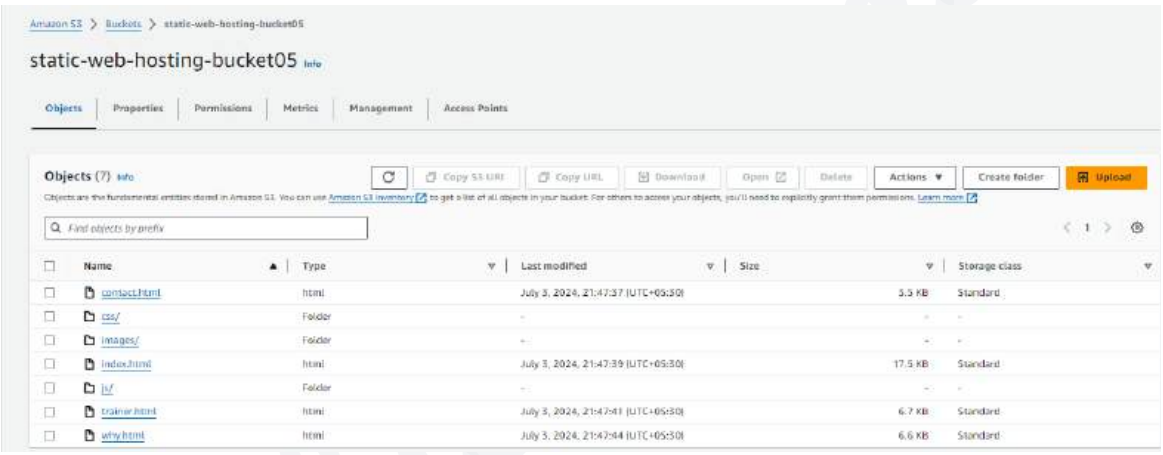


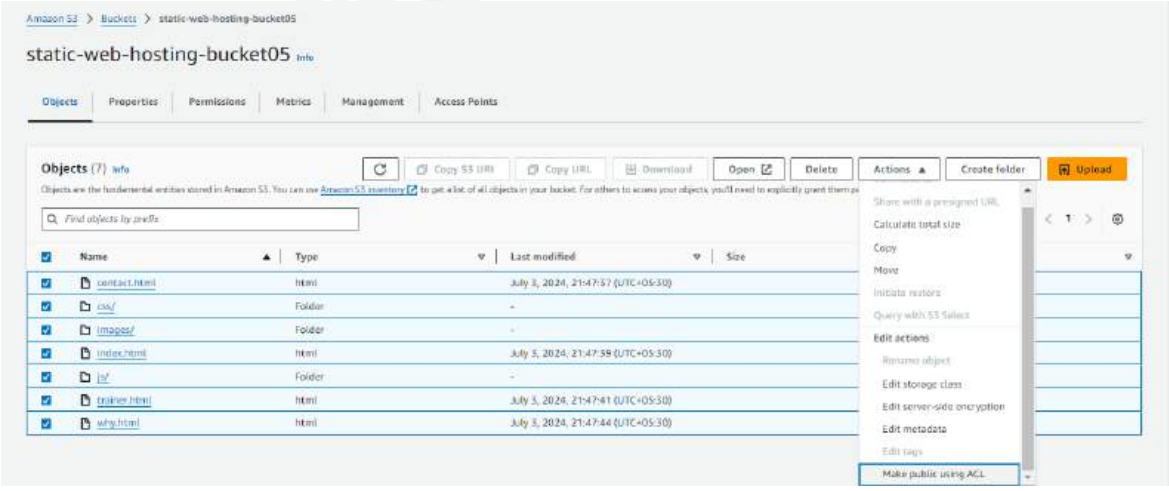






2.12 To make all the Objects in the Bucket **Public,**Now Click on **Objects,** then select all the objects in the bucket, now go to Make Public Using ACL option and then Click on Make Public.





Amazon S3 > Buckets > static-web-hosting-bucket05 > Make public

Make public Info

The make public action enables public read access in the object access control list (ACL) settings. Learn more.

Public access is blocked because Block Public Access settings are turned on for this bucket
To determine which settings are turned on, check your Block Public Access settings for this bucket. Learn more about using Amazon S3 Block Public Access

- When public read access is enabled and not blocked by Block Public Access settings, anyone in the world can access the specified objects.
- This action applies to all objects within the specified folders. Objects added to these folders while the action is in progress might be affected.

- When public read access is enabled and not blocked by Block Public Access settings, anyone in the world can access the specified objects.
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Specified objects

Find objects by name

Name	Type	Last modified	Size
contact.html	html	July 3, 2024, 21:47:37 (UTC+05:30)	5.5 KB
css/	Folder	-	-
images/	Folder	-	-
index.html	html	July 3, 2024, 21:47:39 (UTC+05:30)	17.5 KB
js/	Folder	-	-
trainer.html	html	July 3, 2024, 21:47:41 (UTC+05:30)	6.7 KB
why.html	html	July 3, 2024, 21:47:44 (UTC+05:30)	6.6 KB

Cancel Make public

Public access is blocked because Block Public Access settings are turned on for this bucket
To determine which settings are turned on, check your Block Public Access settings for this bucket. Learn more about using Amazon S3 Block Public Access

Cancel

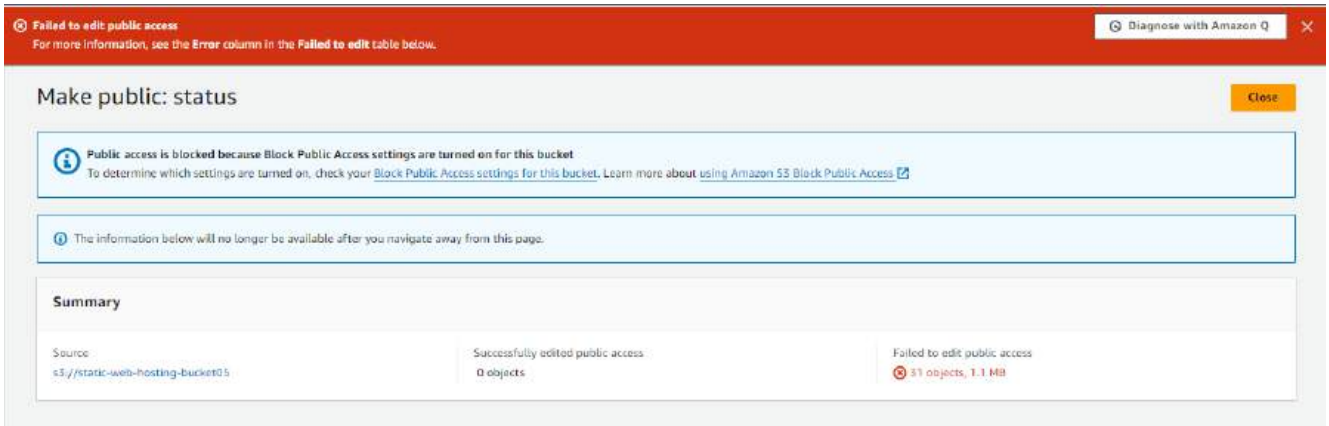
The information below will no longer be available after you navigate away from this page.

Summary

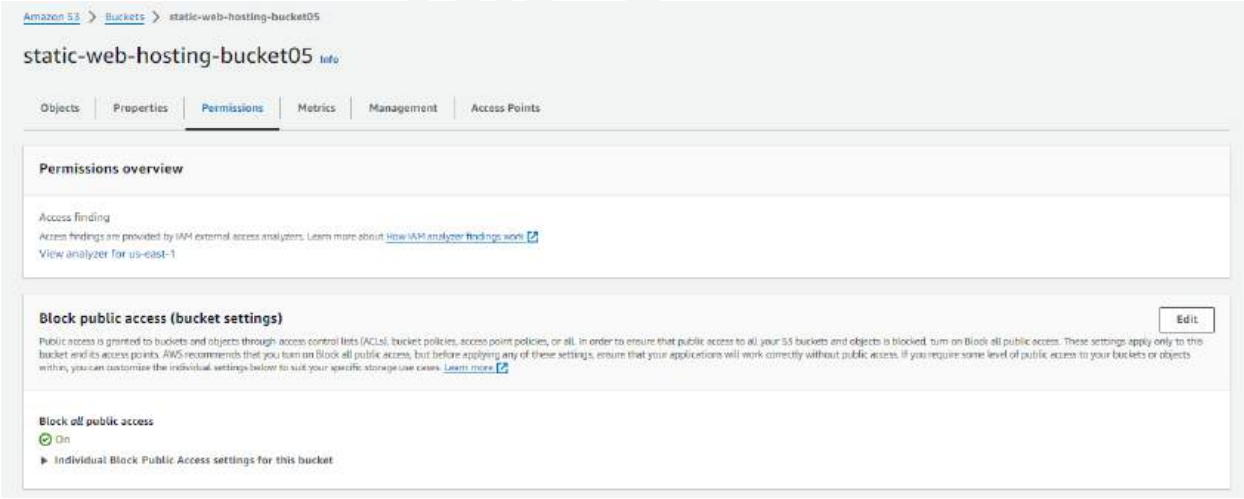
Source	Successfully edited public access	Failed to edit public access
s3://static-web-hosting-bucket05	0 objects	0 objects

Troubleshooting:

- Objects in the Bucket fail to be Public because we have to make the Objects in the Bucket to access Publicly.



- To make the Objects in the Bucket access Publicly, Go to the **Permissions** and the **Block Public Access** and Click on **Edit** Option and then **Uncheck the Block All Public Access** and click on **Save Changes** and then **Confirm**.



Amazon S3 > Buckets > static-web-hosting-bucket05 > Edit Block public access (bucket settings)

Edit Block public access (bucket settings) Info

Block public access (bucket settings)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. Learn more

- [x] **Block *all* public access**
  Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.
  - [x] **Block public access to buckets and objects granted through *new* access control lists (ACLs)**
    S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
  - [x] **Block public access to buckets and objects granted through *any* access control lists (ACLs)**
    S3 will ignore all ACLs that grant public access to buckets and objects.
  - [x] **Block public access to buckets and objects granted through *new* public bucket or access point policies**
    S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
  - [x] **Block public and cross-account access to buckets and objects through *any* public bucket or access point policies**
    S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Cancel | Save changes

Amazon S3 > Buckets > static-web-hosting-bucket05 > Edit Block public access (bucket settings)

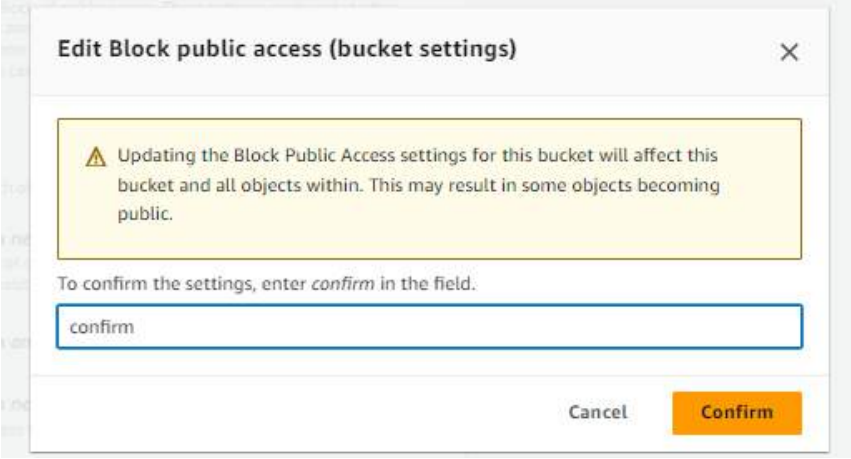
Edit Block public access (bucket settings) Info

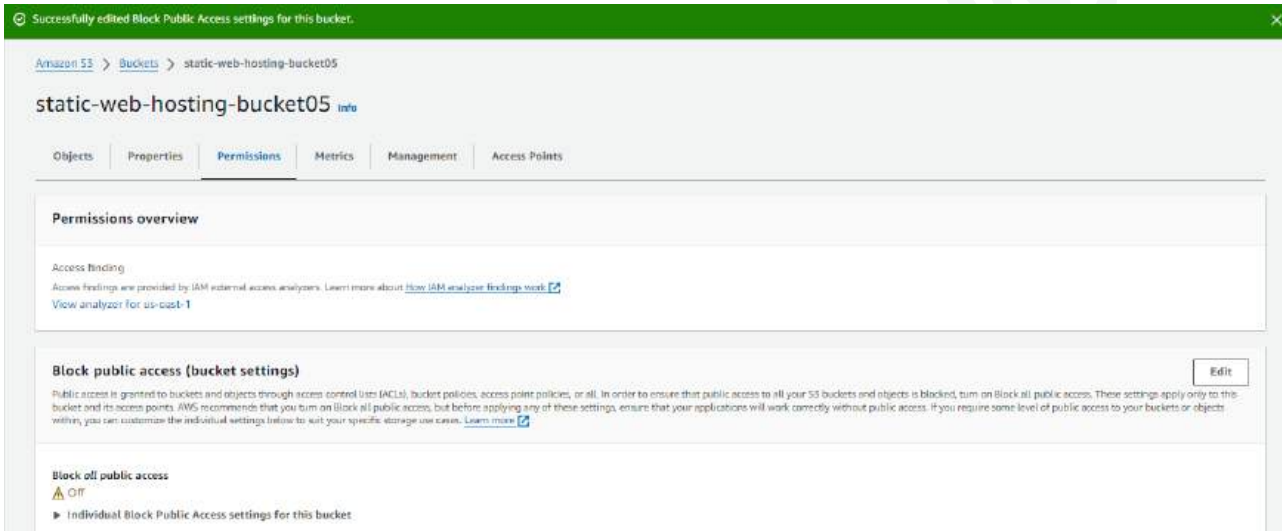
Block public access (bucket settings)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. Learn more

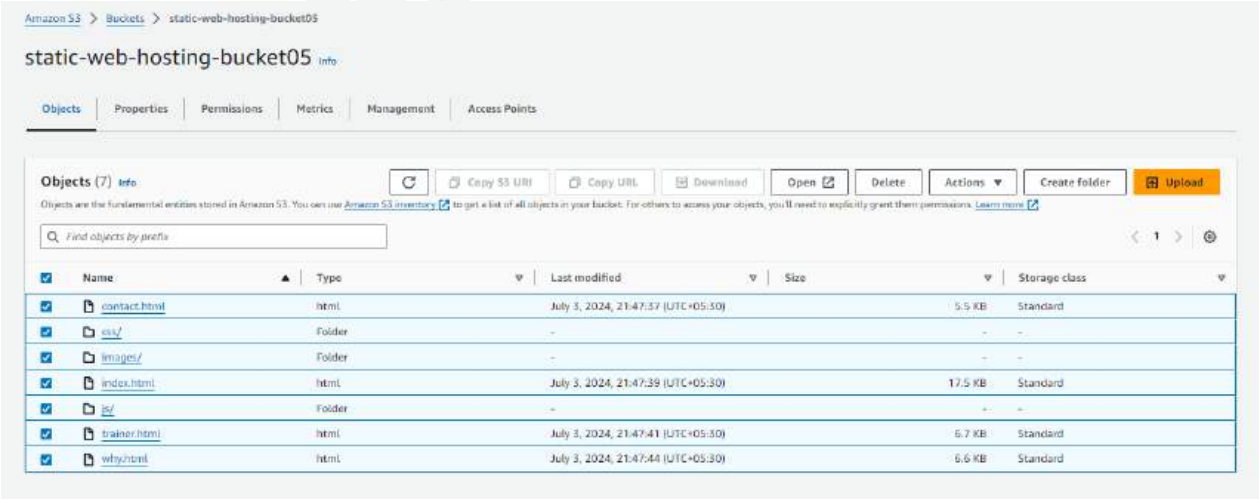
- [ ] **Block *all* public access**
  Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.
  - [ ] **Block public access to buckets and objects granted through *new* access control lists (ACLs)**
    S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
  - [ ] **Block public access to buckets and objects granted through *any* access control lists (ACLs)**
    S3 will ignore all ACLs that grant public access to buckets and objects.
  - [ ] **Block public access to buckets and objects granted through *new* public bucket or access point policies**
    S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
  - [ ] **Block public and cross-account access to buckets and objects through *any* public bucket or access point policies**
    S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

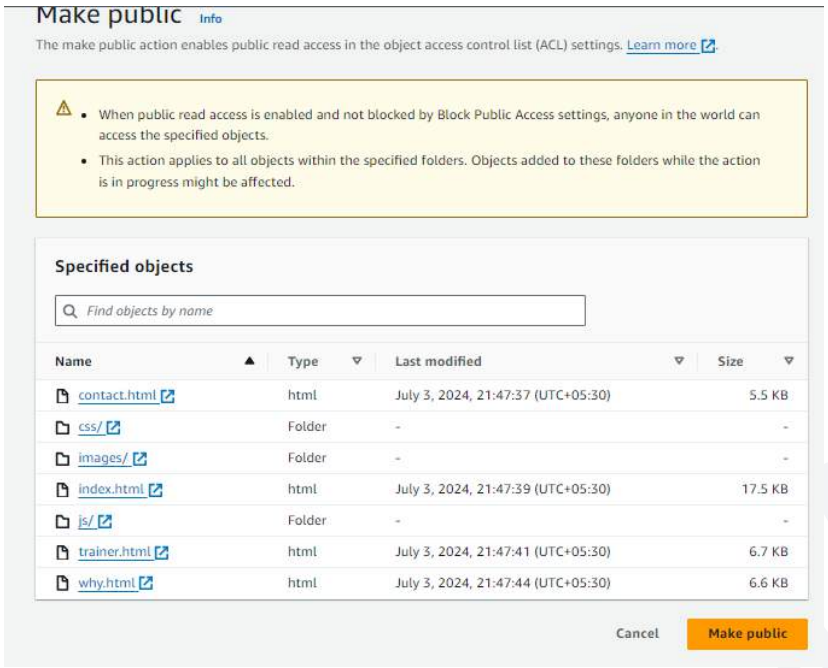
Cancel | Save changes

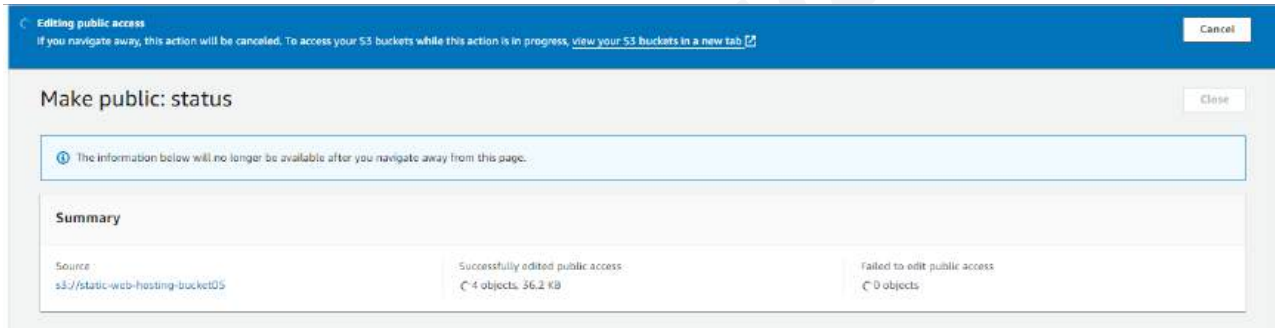




➢ Now we can access the content in the Bucket Publicly.







2.13 Now to check whether the website is publicly accessed or not, Go to **Properties** and scroll down to **Static website Hosting** and then copy the **URL** and now paste on the web browser's URL.

Amazon S3 > Buckets > static-web-hosting-bucket05

static-web-hosting-bucket05 Info

Objects | Properties | Permissions | Metrics | Management | Access Points

Bucket overview

AWS Region	Amazon Resource Name (ARN)	Creation date
US East (N. Virginia) us-east-1	arn:aws:s3:::static-web-hosting-bucket05	July 3, 2024, 21:39:05 (UTC+05:30)

Bucket Versioning

Edit

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. Learn more

Bucket Versioning
Disabled

Multi-factor authentication (MFA) delete
An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. Learn more
Disabled

Static website hosting

Edit

Use this bucket to host a website or redirect requests. Learn more

Static website hosting
Enabled

Hosting type
Bucket hosting

ebsite endpoint copied

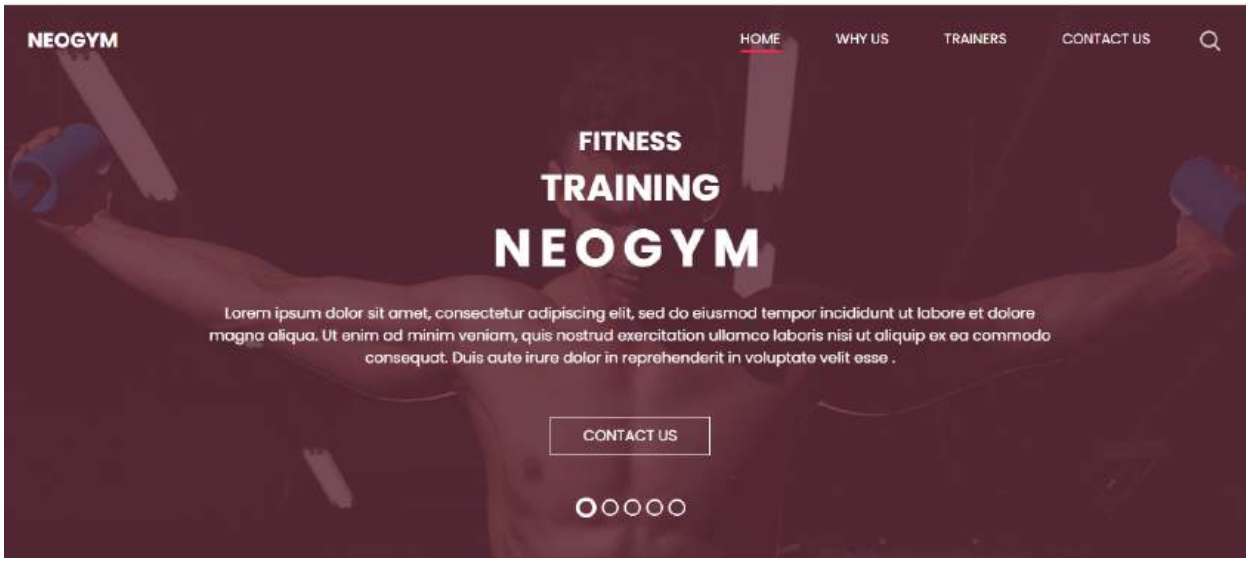
int

bucket as a static website, the website is available at the AWS Region-specific website endpoint of the bucket. Learn more

http://static-web-hosting-bucket05.s3-website-us-east-1.amazonaws.com

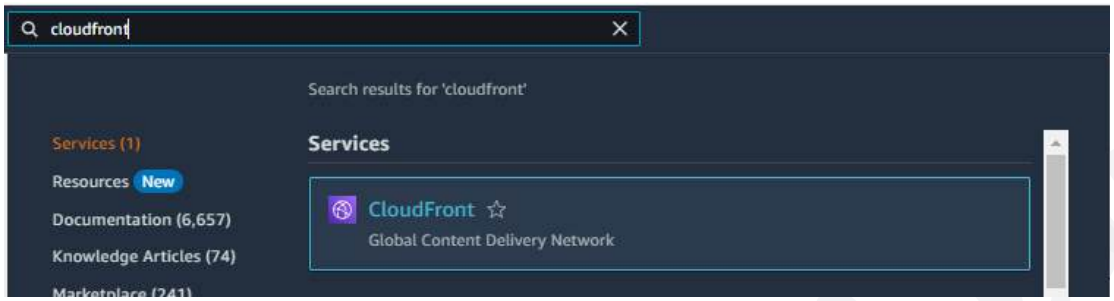
Not secure static-web-hosting-bucket05.s3-website-us-east-1.amazonaws.com

ChatGPT | (78) YouTube | Manage AWS Resou... | Online Courses - Le... | Sign in to GitHub · ... | Google Meet | draw.io | WhatsApp | Medium | Login | | LinkedIn | Home | Myneuki



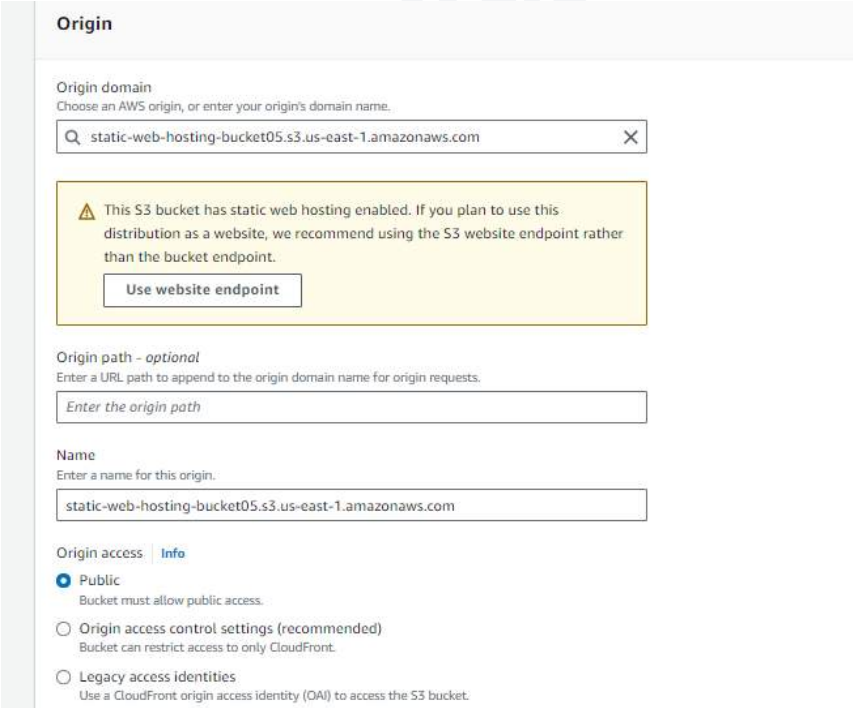
Step 3: Launch CloudFront for distribution of web content

3.1 In Dashboard, search for **Cloudfront service**, click CloudFront to open it.

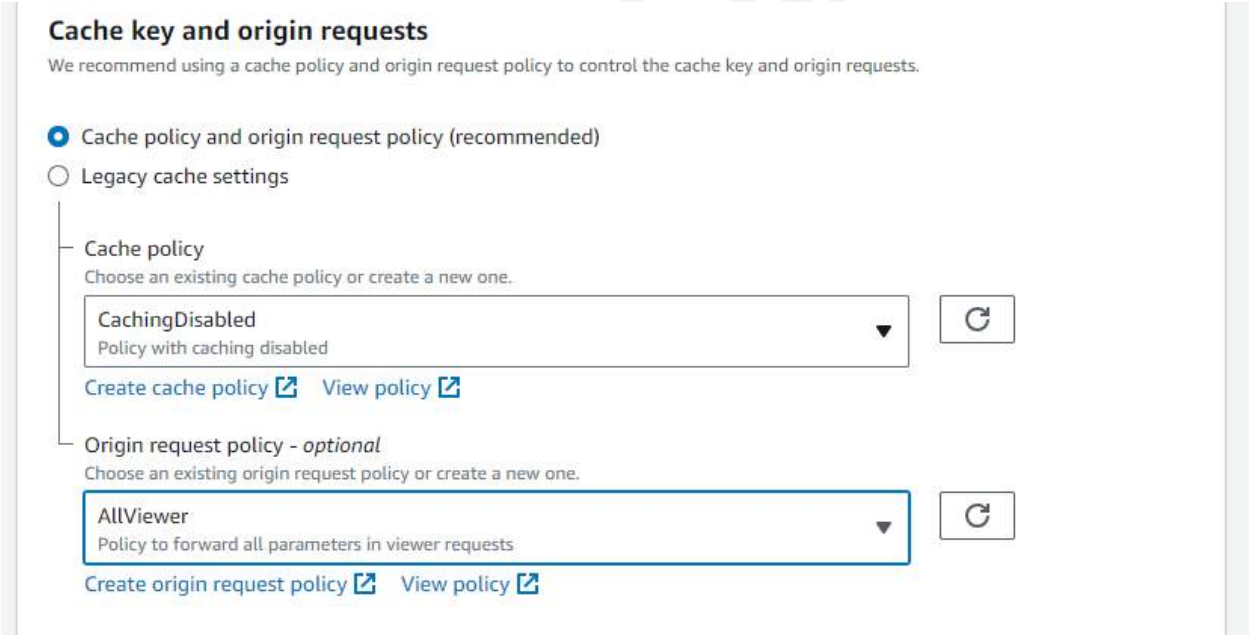


3.2 Click on **create a cloudfront distribution** to configure.

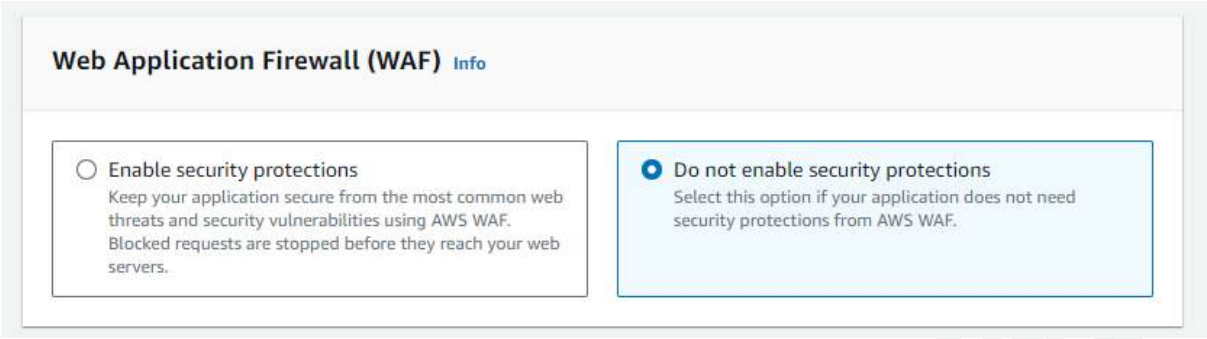
3.3 Under the **origin domain** select the Amazon S3, which was created prior.



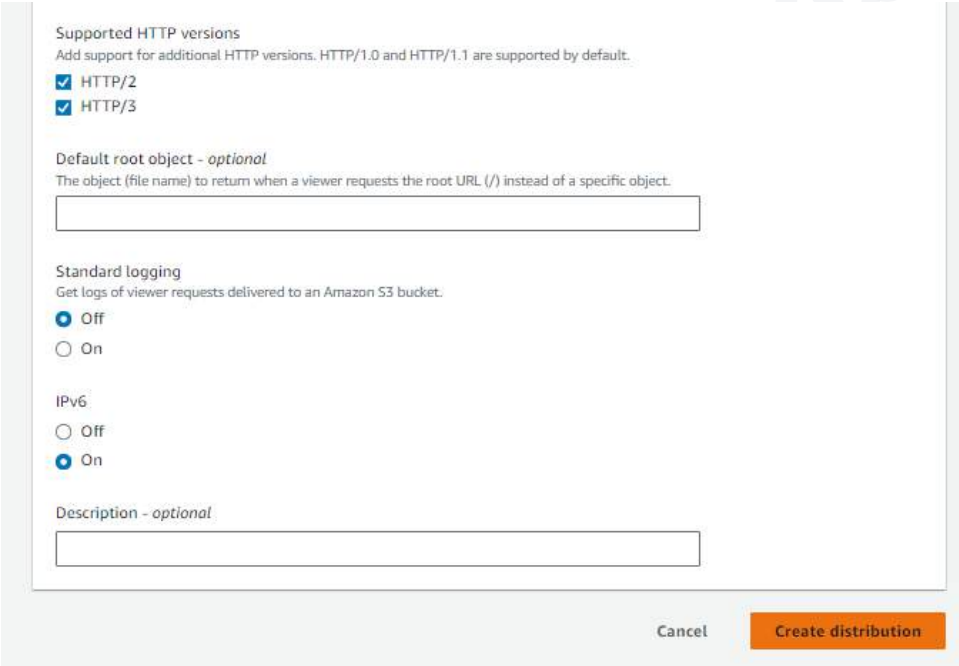
3.4 Now, scroll down further to **Cache key and origin requests** here CachePolicy make it as **CachingDisabled** and Origin request policy as **AllViewer**.

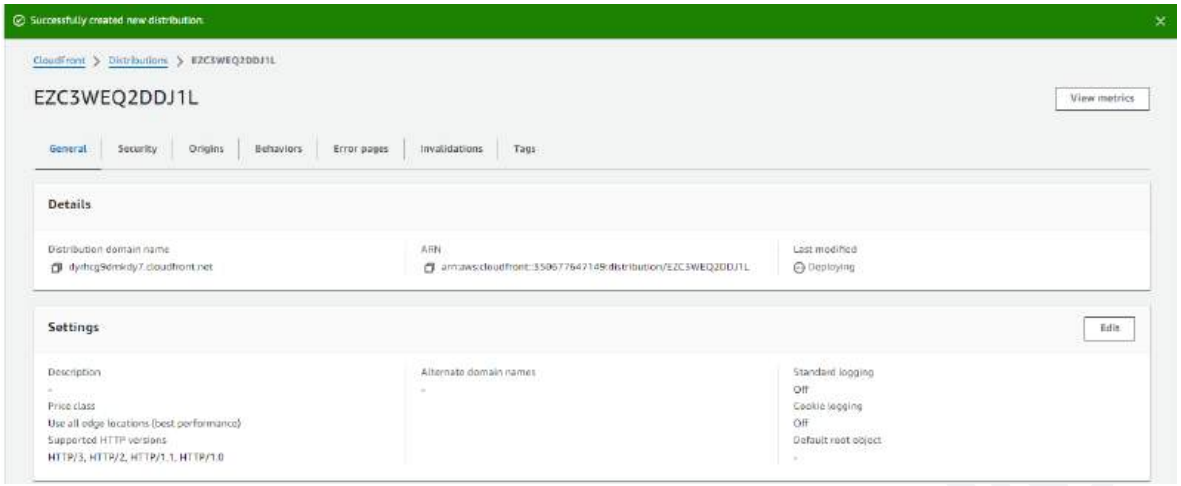


3.5 Disable the WAF by selecting **Do not enable security protections** and rest leave as default and click **Create distribution.**



3.6 click on **Create distribution** so that CloudFront Distribution will be creating and Successfully **created CloudFront new distribution**.

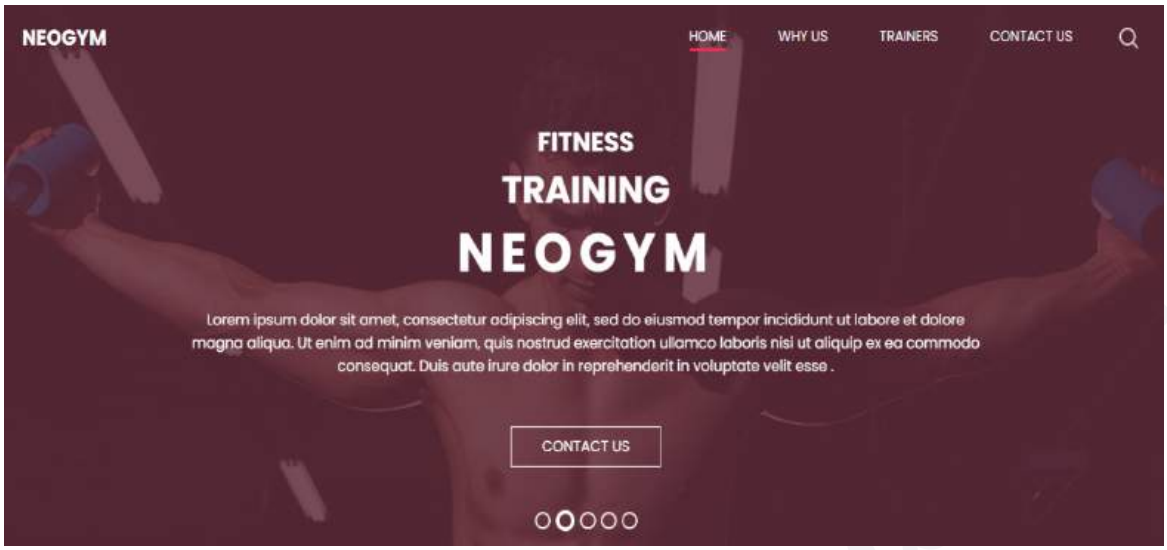




Bucket Policy:

```
JSON
Bucket Policy:
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "PublicReadGetObject",
            "Effect": "Allow",
            "Principal": "*",
            "Action": [
                "s3:GetObject"
            ],
            "Resource": [
                "arn:aws:s3:::Bucket-Name/*" (Replace with your bucket-name)
            ]
        }
    ]
}
```

3.7 Now copy the **Distribution domain name** and paste it in a new tab to run the frontend website.



Result 2:

Task on to launch a Static website using Amazon S3 (Simple Storage Service) via Amazon CloudFront (Content Delivery Network) successfully accomplished.

Conclusion:

Case 1:

Static Web Hosting Without Using Amazon S3 and Using CloudFront

Alternative Storage:

- Instead of S3, you can use other origins for your static content, such as on-premises servers, EC2 instances.

Management:

- You have to manage and scale these storage solutions yourself, which can be more complex compared to using S3.

Amazon CloudFront:

- **CDN:** CloudFront still acts as the CDN, caching content at edge locations for low-latency delivery.
- **Configuration:** You configure CloudFront to pull content from your specified origin (e.g., a web server or another cloud storage service).

Case 2:

Amazon S3 (Simple Storage Service):

- **Storage:** Amazon S3 is used to store the static files (HTML, CSS, JavaScript, images, etc.).
- **Scalability:** S3 scales automatically to handle large amounts of data and many requests.
- **Cost:** You pay for the storage used and the data transferred out of S3.
- **Accessibility:** S3 buckets can be configured to serve public content directly over HTTP/HTTPS.

Amazon CloudFront:

- **CDN:** CloudFront is a Content Delivery Network (CDN) that caches content at edge locations around the world, reducing latency by serving content closer to the user.
- **Integration with S3:** When integrated with S3, CloudFront can automatically pull content from S3 and distribute it globally.
- **Security:** CloudFront provides additional security features such as DDoS protection, SSL/TLS encryption, and access control.

Comparison:

Ease of Use:

- **S3 and CloudFront:** Easier to set up and manage because S3 and CloudFront are both AWS services, and they integrate seamlessly.

Rakesh Lyadalla

- **Other Origins and CloudFront:** More complex setup and management, especially if using non-AWS storage solutions.

Cost:

- **S3 and CloudFront:** Costs are predictable and typically lower due to AWS's integrated pricing models.
- **Other Origins and CloudFront:** Costs can vary depending on the origin storage solution and potential additional charges for data transfer between services.

Performance:

- **S3 and CloudFront:** Optimized for performance due to AWS's global infrastructure and direct integration.
- **Other Origins and CloudFront:** Performance can be good but may require additional tuning and optimization.

Security:

- **S3 and CloudFront:** Robust security features with AWS IAM, bucket policies, and CloudFront security integrations.
- **Other Origins and CloudFront:** Security must be configured and managed separately for the origin storage solution.

- Using Amazon S3 with CloudFront is generally more straightforward and integrated, making it a popular choice for static web hosting. However, using CloudFront with other origins provides flexibility but requires more management and potentially higher complexity.

Rohith Gyadalla